

Comparative Acoustical Analysis: Impact of Noise Criteria (NC) on Patient Ward Intelligibility and Comfort



BSTC- 582 Acoustical Modeling
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Noise Criteria (NC): Technical Definition and Clinical Significance

Noise Criteria (NC) is a single-number rating system used to quantify the continuous background noise floor in indoor environments, specifically targeting the sound spectrum produced by HVAC systems and mechanical equipment. It is determined by plotting octave-band sound pressure levels against standardized NC curves, where the rating corresponds to the lowest curve not exceeded by any spectrum point.

In healthcare environments, specifically Patient Wards, the NC level is a critical performance metric. High background noise levels (exceeding NC 35) are directly correlated with:

- **Patient Recovery:** Disruption of sleep cycles, increased heart rate, and elevated stress levels, which impede physiological healing.
- **Speech Intelligibility:** High noise floors mask the acoustic transients of speech, increasing the cognitive load on medical staff and heightening the risk of communication errors during critical care.
- **Acoustic Privacy:** Conversely, an excessively low NC (below NC 25) can compromise patient-doctor confidentiality by failing to provide adequate sound masking between adjacent ward areas.

1. Introduction

The acoustical environment of a healthcare facility is a fundamental component of patient care and clinical safety. This report investigates the impact of background Noise Criteria (NC) on the acoustical performance of a specific hospital ward layout.

The primary objective is to evaluate how a 5 dB increase in HVAC background noise—transitioning from NC 30 to NC 35 affects speech intelligibility and clarity within patient recovery zones.

Excessive background noise in these environments is clinically linked to sleep disruption, elevated patient stress levels, and an increased risk of communication errors among medical staff.

Technical Parameters (500Hz – 2000Hz)

To evaluate the ward's performance, we analyze four distinct metrics across the "Speech Core" (500Hz, 1000Hz, and 2000Hz). These frequencies are where the human ear is most sensitive and where medical alarms are typically centered.

1. EDT (Early Decay Time)

- Definition: The time it takes for the initial 10 dB of sound energy to decay, multiplied by a factor of 6 to normalize it to a 60 dB range.
- Significance: Unlike T30, EDT is highly dependent on the location of the listener and the nearby reflective surfaces (walls/ceilings). It dictates the subjective "clarity" or "muddiness" a person perceives in the immediate sound field.

2. T30 (Reverberation Time)

- Definition: The time required for the sound pressure level to drop by 60 dB, calculated based on the decay between -5 dB and -35 dB.
- Significance: This is the global "echo" of the room. A high T30 in a 2173 m³ ward causes "noise buildup," where background sounds (trolleys, footsteps) linger too long and wash out clear communication.

3. C50 (Clarity)

- Definition: The logarithmic ratio (in dB) of the "Early" sound energy (received within the first 50ms) to the "Late" sound energy (received after 50ms).
- Significance: In a ward, the first 50ms of a sound carries the useful information (consonants). If the late energy (reflections) is too high, C50 drops, making speech sound blurred or unintelligible.

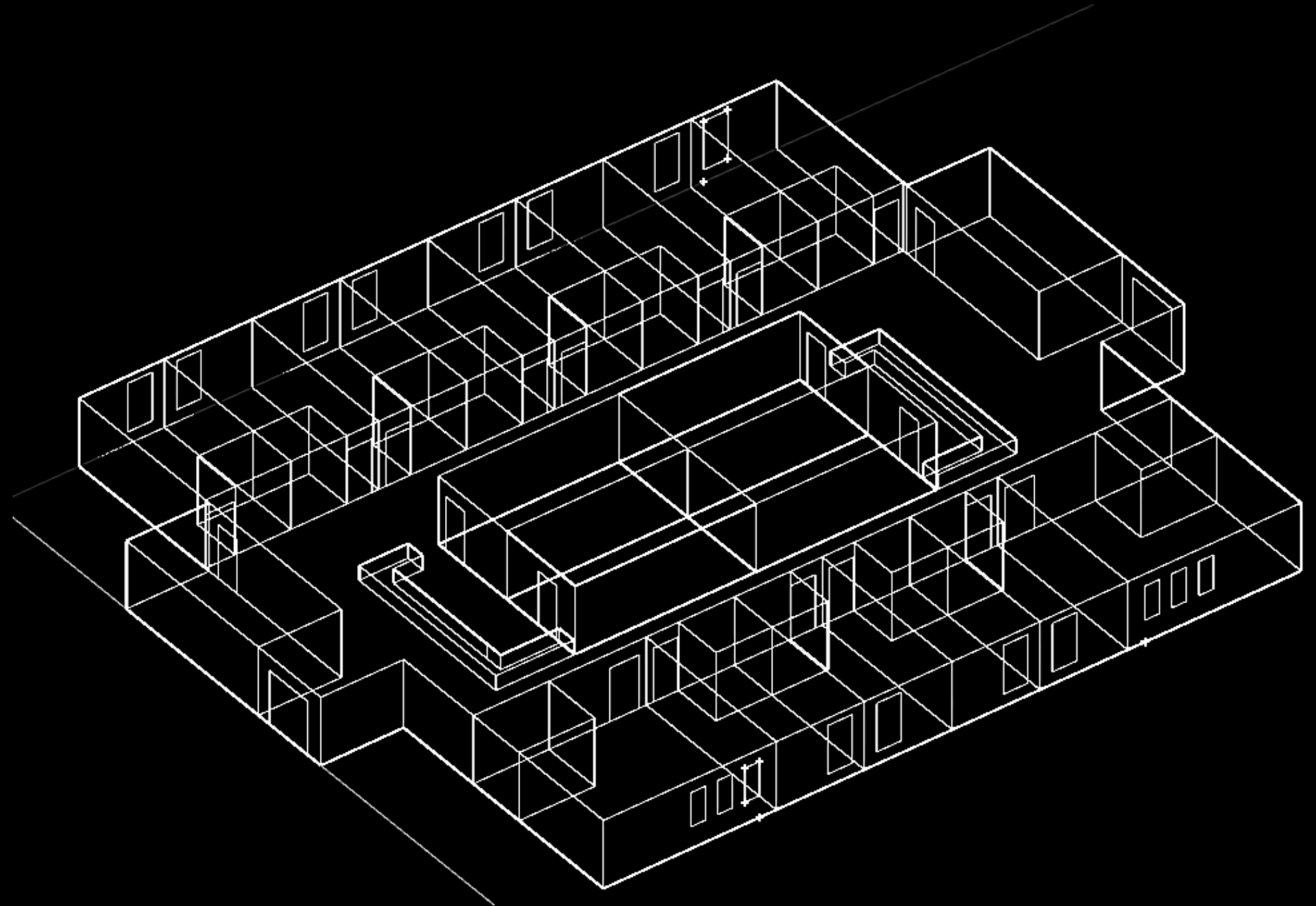
4. STI (Speech Transmission Index)

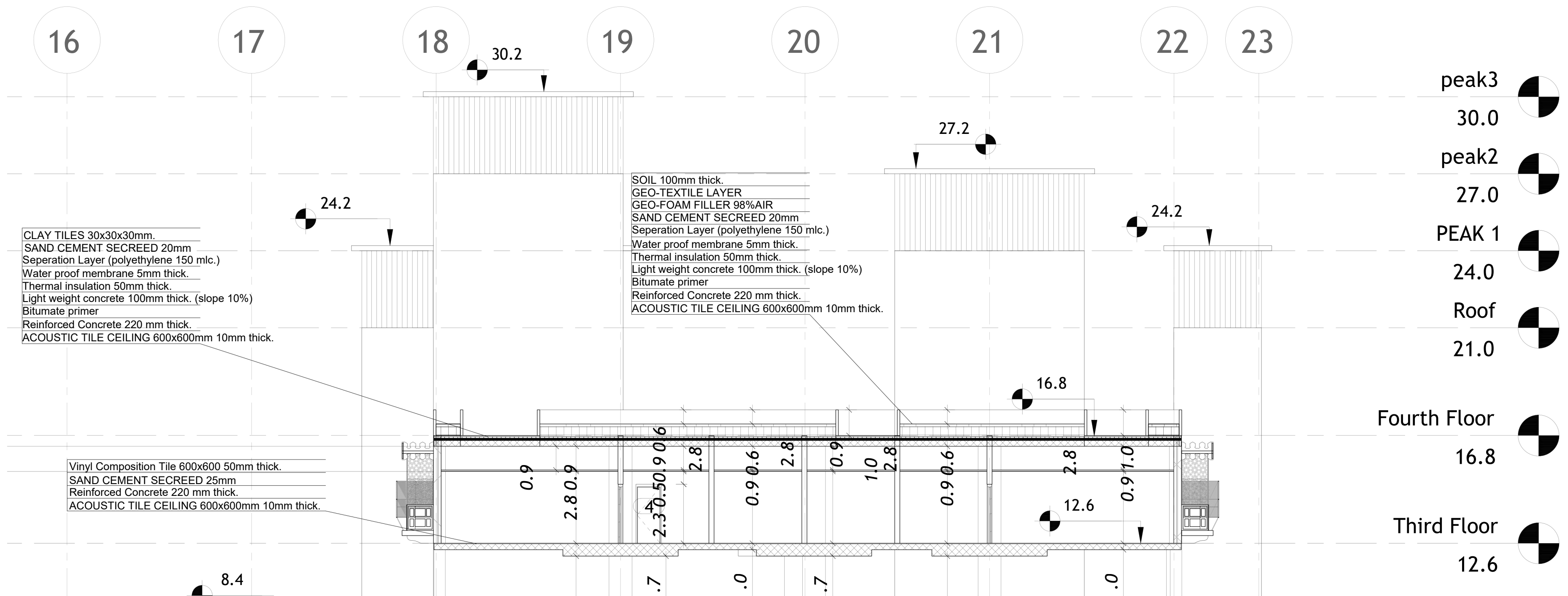
- Definition: A numeric representation of speech quality on a scale of 0 (unintelligible) to 1.0 (perfect).
- The NC Effect: STI is the only metric here that directly accounts for Background Noise (NC). As you shift from NC 30 to NC 35, the noise "masks" the speech signal, lowering the STI even if the room's physical reverberation remains the same.

Methodology & Simulation Setup

1. Spatial Configuration & Architectural Data

- Total Estimated Volume: 2173.00 m³.
- Clear Height: 2.8 meters, a standard clinical height that influences early ceiling reflections.
- Room Distribution: * 12 Mid-Size Ward Rooms for standard patient recovery.
 - 2 Large Ward Rooms for high-occupancy.
 - 12 Mid-Size Ward Rooms for standard patient recovery.
- Clinical Hubs: Two central Nurse Stations positioned for maximum oversight.
- Support Zones: 4 Patient Service Rooms (meal prep and utility) located centrally between the nurse stations.



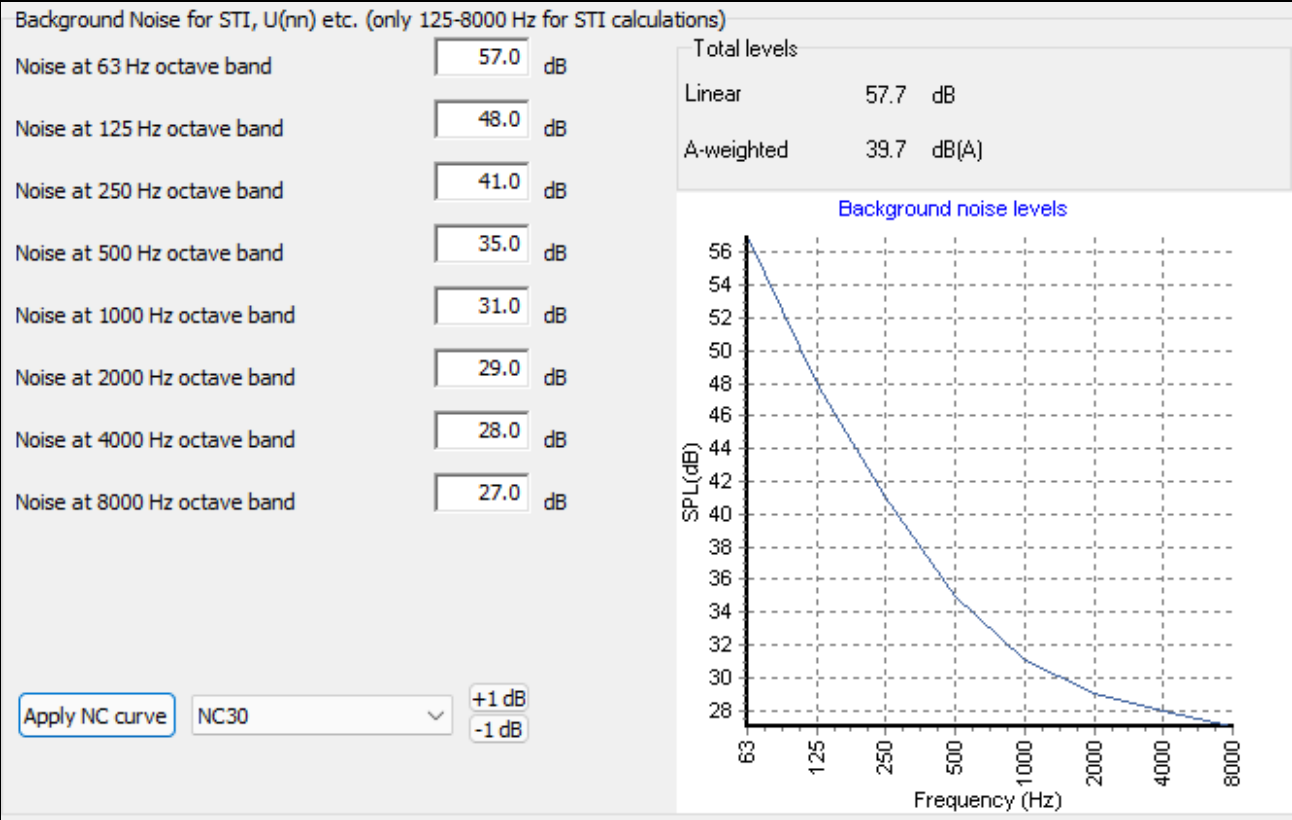


Methodology & Simulation Setup

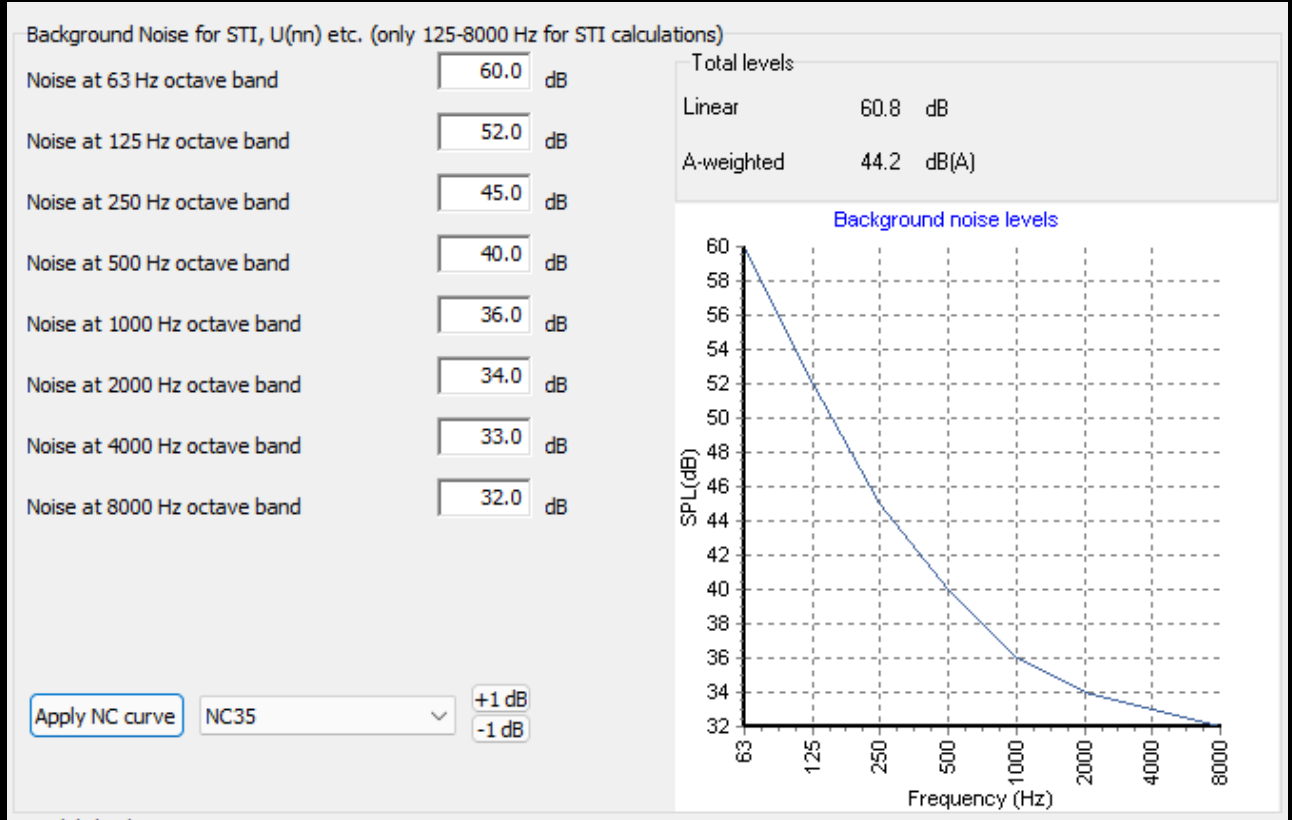
2. Simulation Parameters

- **Software Platform: ODEON Room Acoustics** (utilizing 48,359 rays for precision).
- **Receiver Configuration:** The receiver grid is set at a 1.65m height, simulating a patient sitting on a bed.
- **Comparative Scenarios:**
 - **NC 30 (High Performance):** Total linear level of 57.7 dB (39.7 dBA).
 - **NC 35 (Standard Target):** Total linear level of 60.8 dB (44.2 dBA).
- **Frequency Analysis:** Noise levels will be analysed across the 500-1000-2000 Hz octave bands for comprehensive STI and intelligibility calculations.

NC 30 Curve



NC 35 Curve

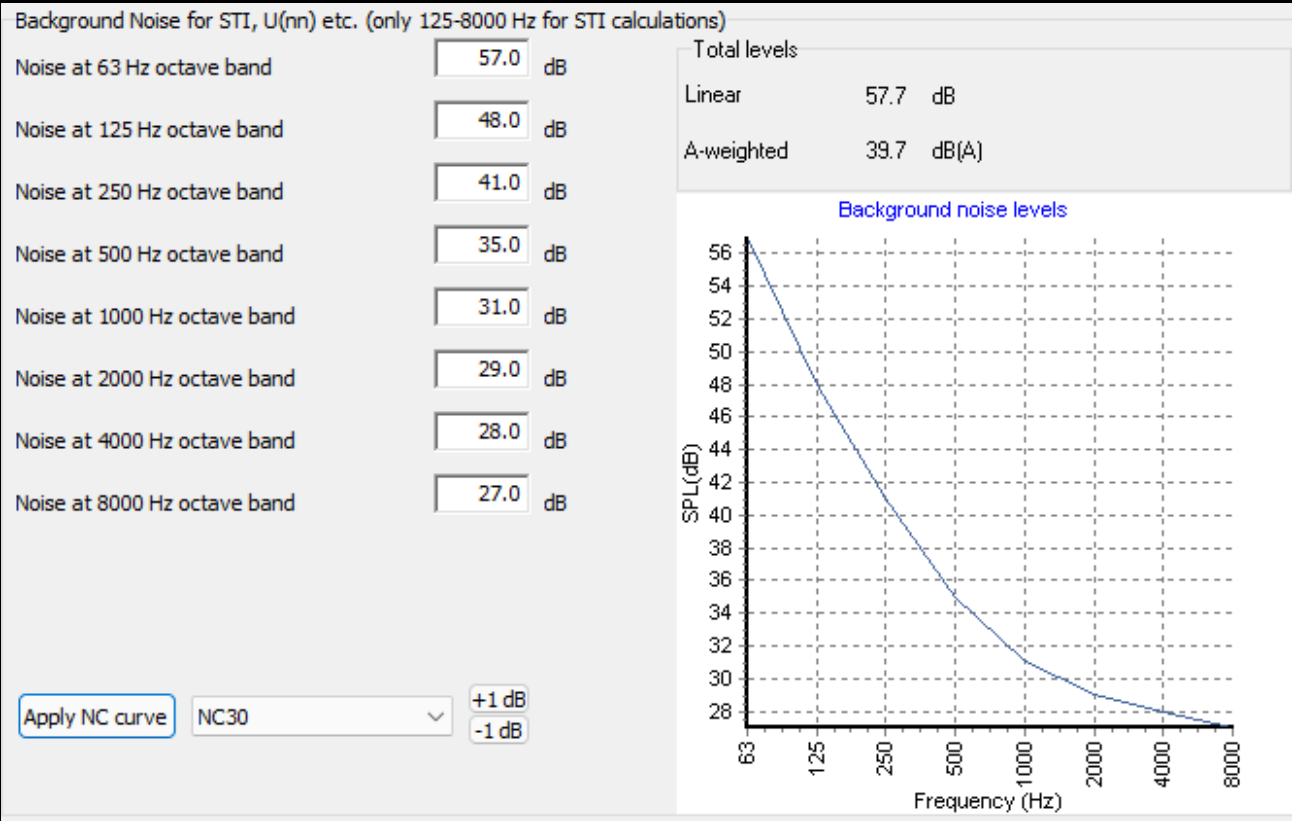


Methodology & Simulation Setup

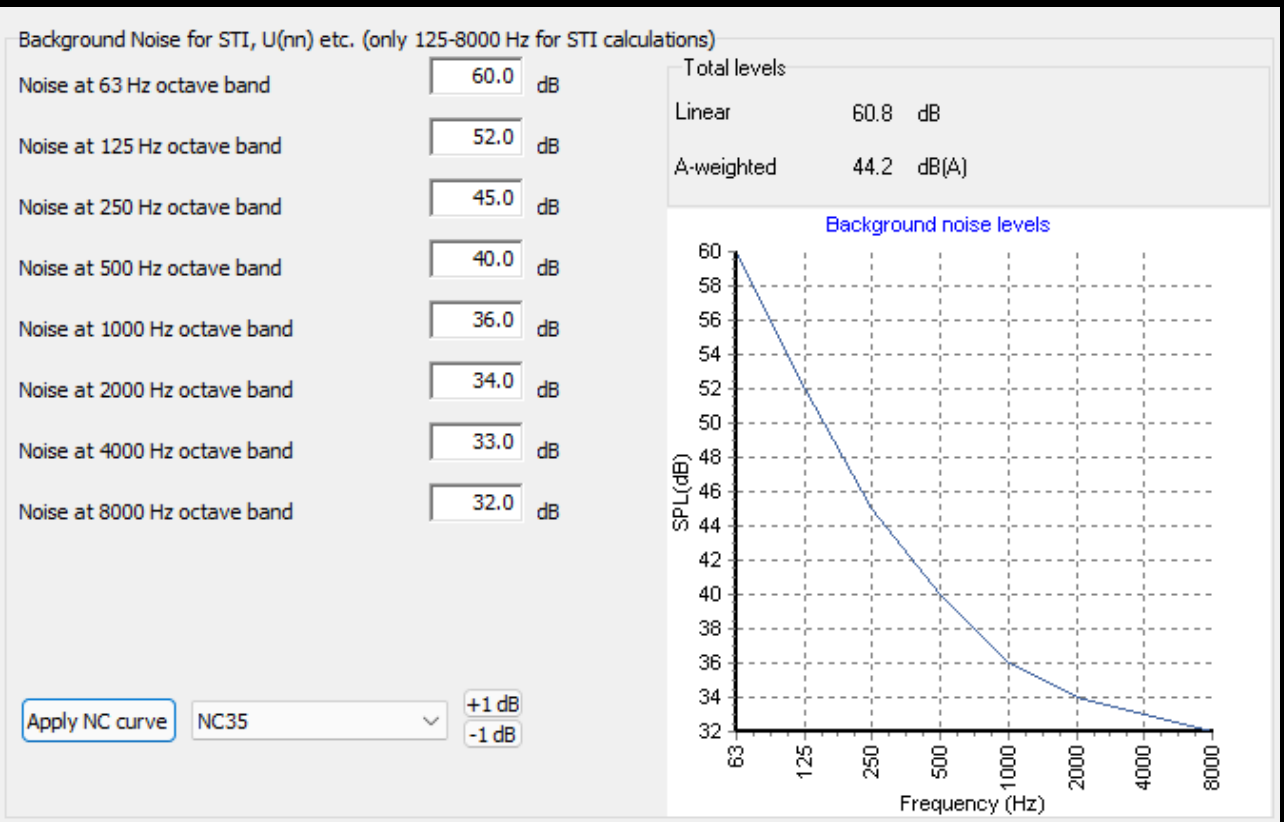
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NC 30 Curve



NC 35 Curve



Materials

Plaster Finished Walls

	57	4002	0.050	0.000	Normal ▼	1-WALLS		
Mat. 4002: Painted plaster surface (Kristensen, 1984)								
	63 Hz	125 Hz	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
	0.02000	0.02000	0.02000	0.02000	0.02000	0.02000	0.02000	0.02000

Acoustic Tiles Ceiling

	59	12200	0.050	0.000	Normal	▼	CEILING	
Mat. 12200: (Example 12200 12299) Ecophon (href= "http://www.ecophon.com/")								
	63 Hz	125 Hz	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
	0.15000	0.15000	0.50000	0.95000	1.00000	0.95000	0.85000	0.65000

Vinyl Tiles Floor

84	6000	0.050	0.000	Normal	▼	FLOOR	
Mat. 6000: Linoleum or vinyl stuck to concrete (Petersen, 1983)							
63 Hz	125 Hz	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
0.02000	0.02000	0.02000	0.03000	0.04000	0.04000	0.05000	0.05000

Wooden Desk-Nurse Station

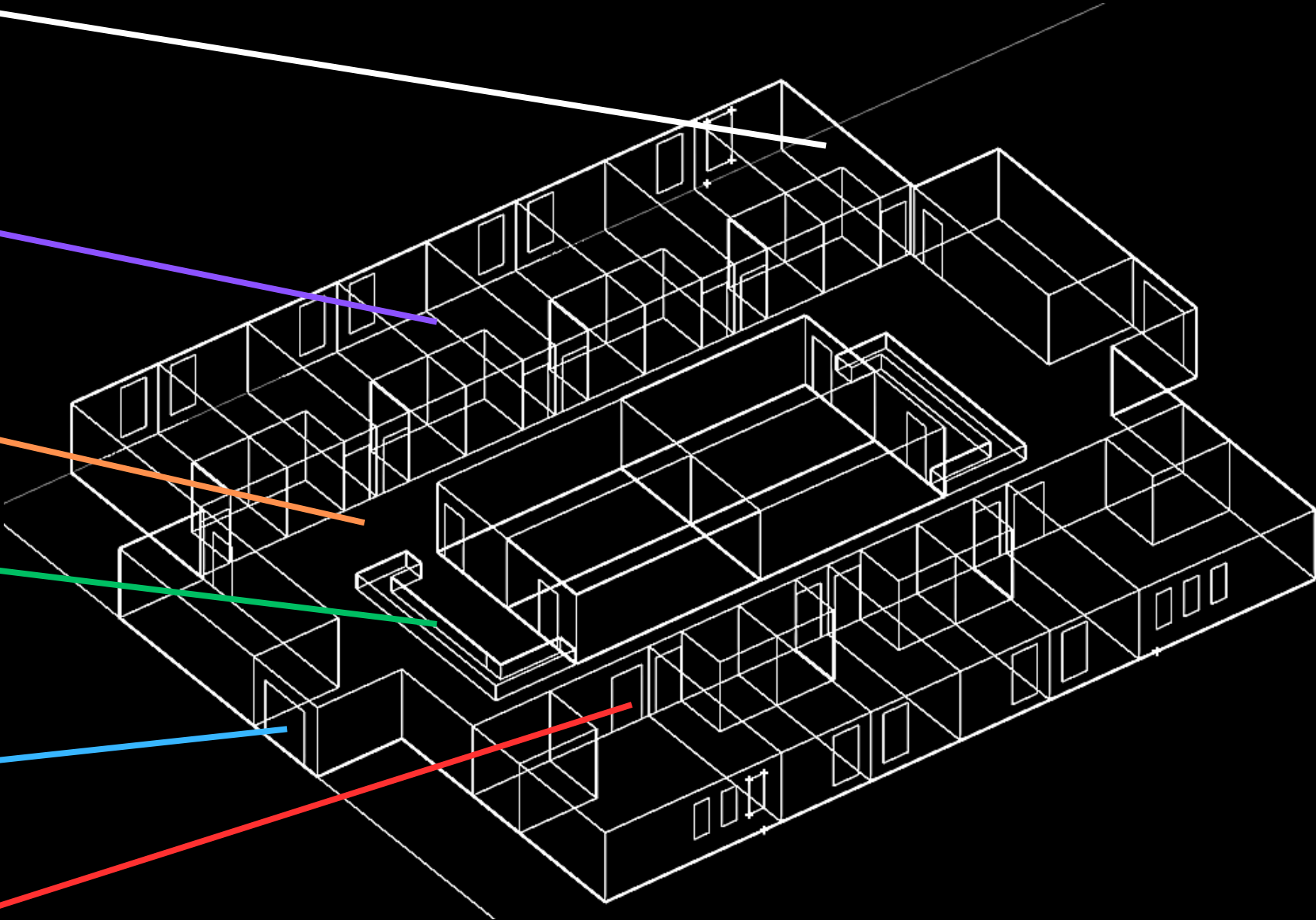
	102	10100	0.050	0.000	Normal	▼	3-NURSE_STATION	
Mat. 10100: (Example 10100-10199) S-LINE Office screen. wood Strict Line (href= "http://www.								
	63 Hz	125 Hz	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
	0.02000	0.02000	0.20000	0.70000	1.00000	0.90000	0.70000	0.70000

Wooden Doors

	77	10007	0.050	0.000	Normal	▼	2-DOORS	
Mat. 10007: Solid wooden door (Bobran, 1973)								
	63 Hz	125 Hz	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
	0.14000	0.14000	0.10000	0.06000	0.08000	0.10000	0.10000	0.10000

Wooden Doors- Transparent For Analysis

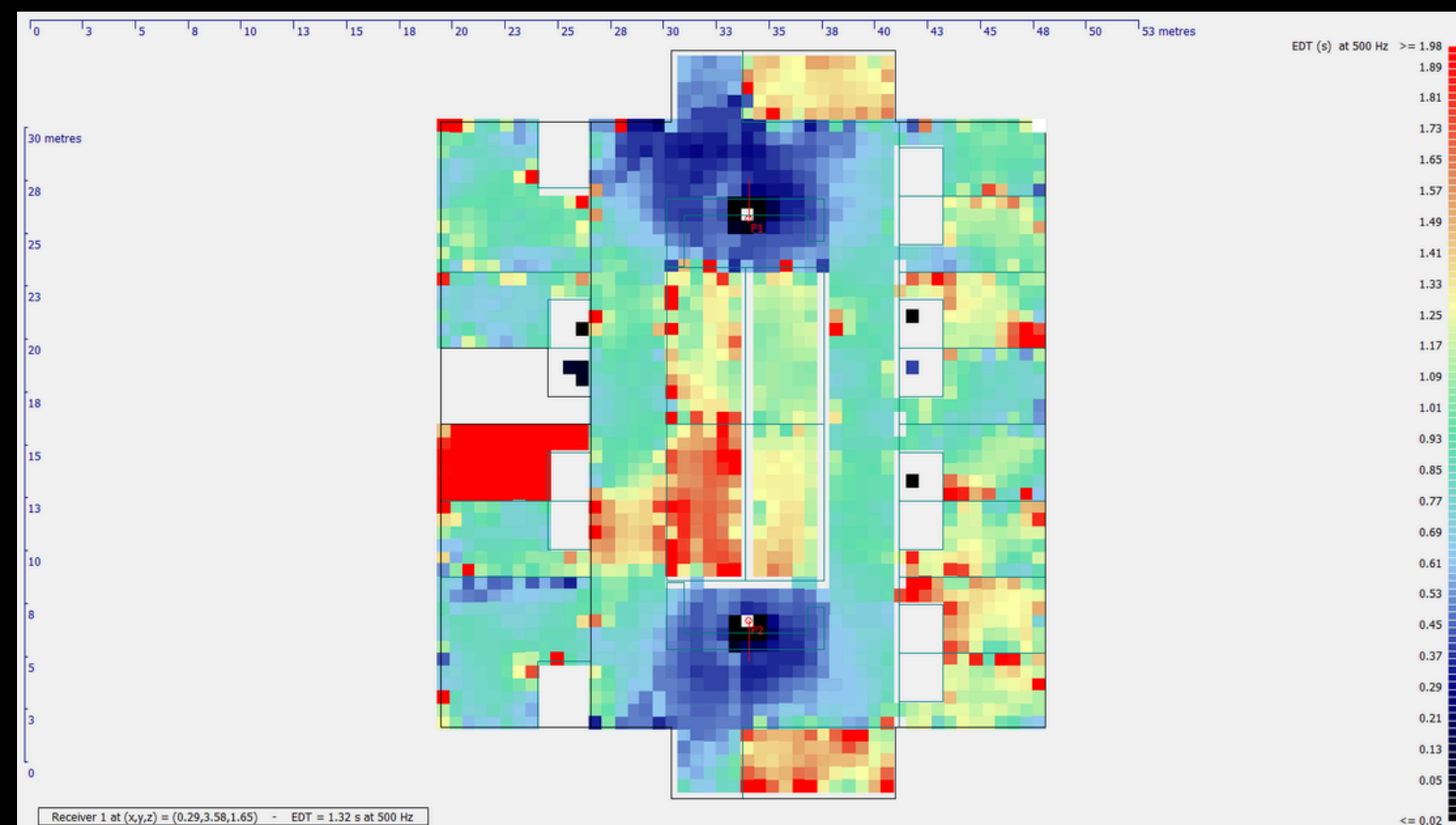
77	10007	0.050	0.000	Normal ▼	2-DOORS		
Mat. 0: Transperant							
63 Hz	125 Hz	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000



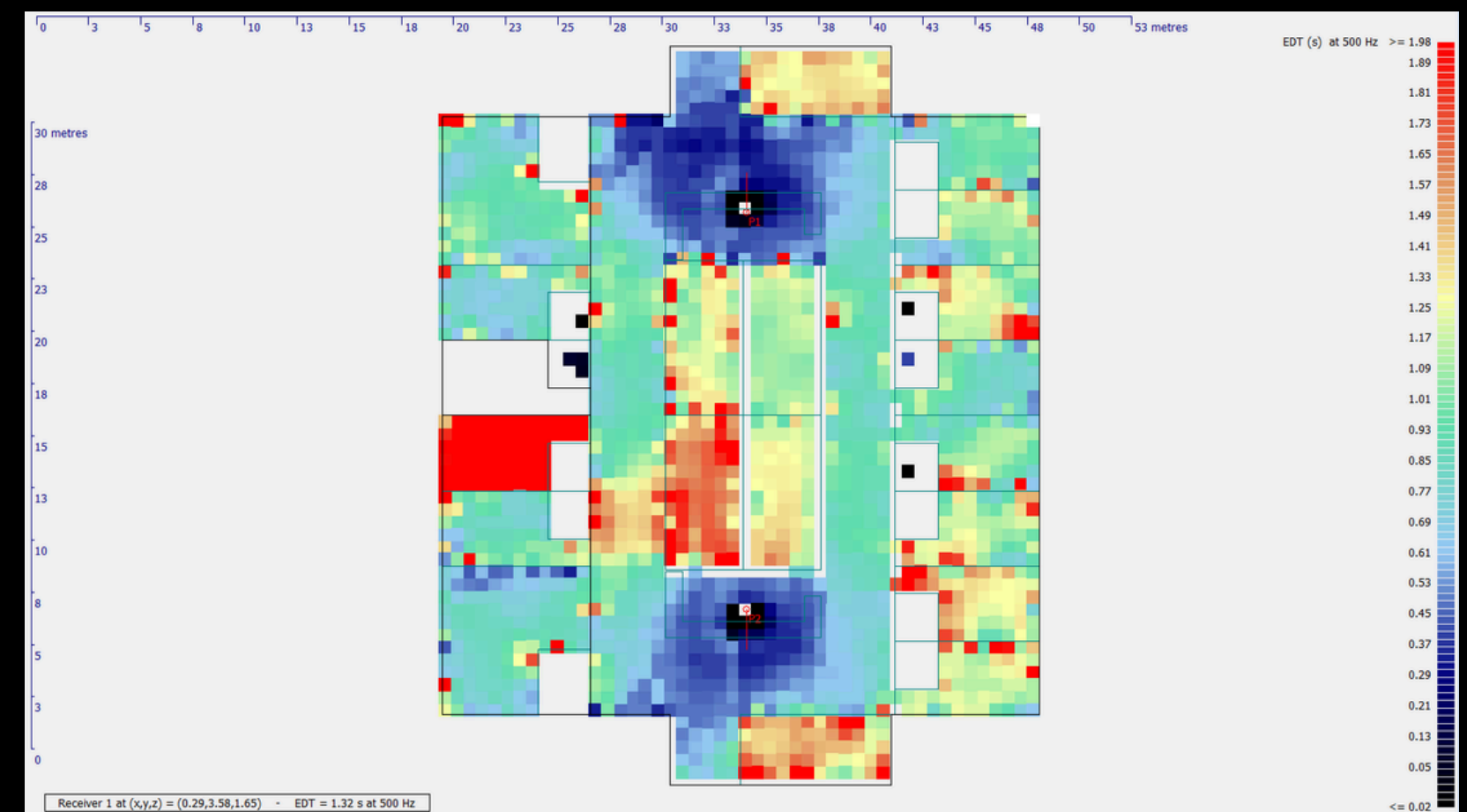
Edt @500 Hz

The Early Decay Time (EDT) at 500 Hz remains identical at 1.32 s for both the NC 30 and NC 35 scenarios, as this parameter is a physical attribute of the room's geometry and material composition—specifically the 2.8 m clear height, plaster walls, and acoustic ceiling tiles—rather than the background noise level. Spatially, the nurse stations show the highest clarity with EDT values below 0.45 s, while a specific mid-size ward on the left indicates a "red zone" of high reverberance exceeding 1.8 s. For a patient at Receiver 1, sitting at a 1.65 m height, the measured EDT of 1.32 s is relatively high for a clinical ward. This suggests that while the physical decay does not change with the HVAC noise floor, the existing reflections at 500 Hz are already significant enough to potentially blur speech.

NC 30



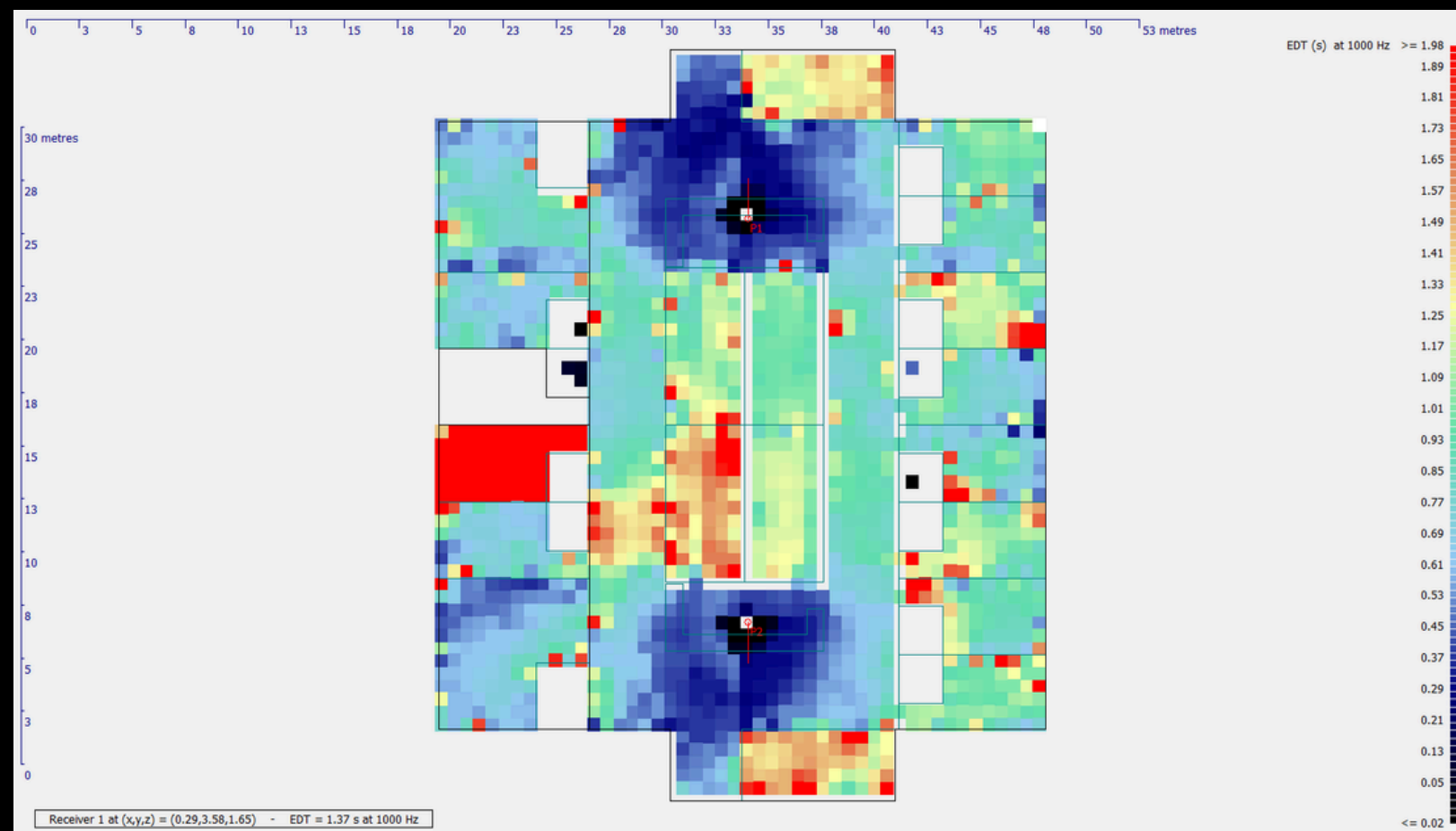
NC 35



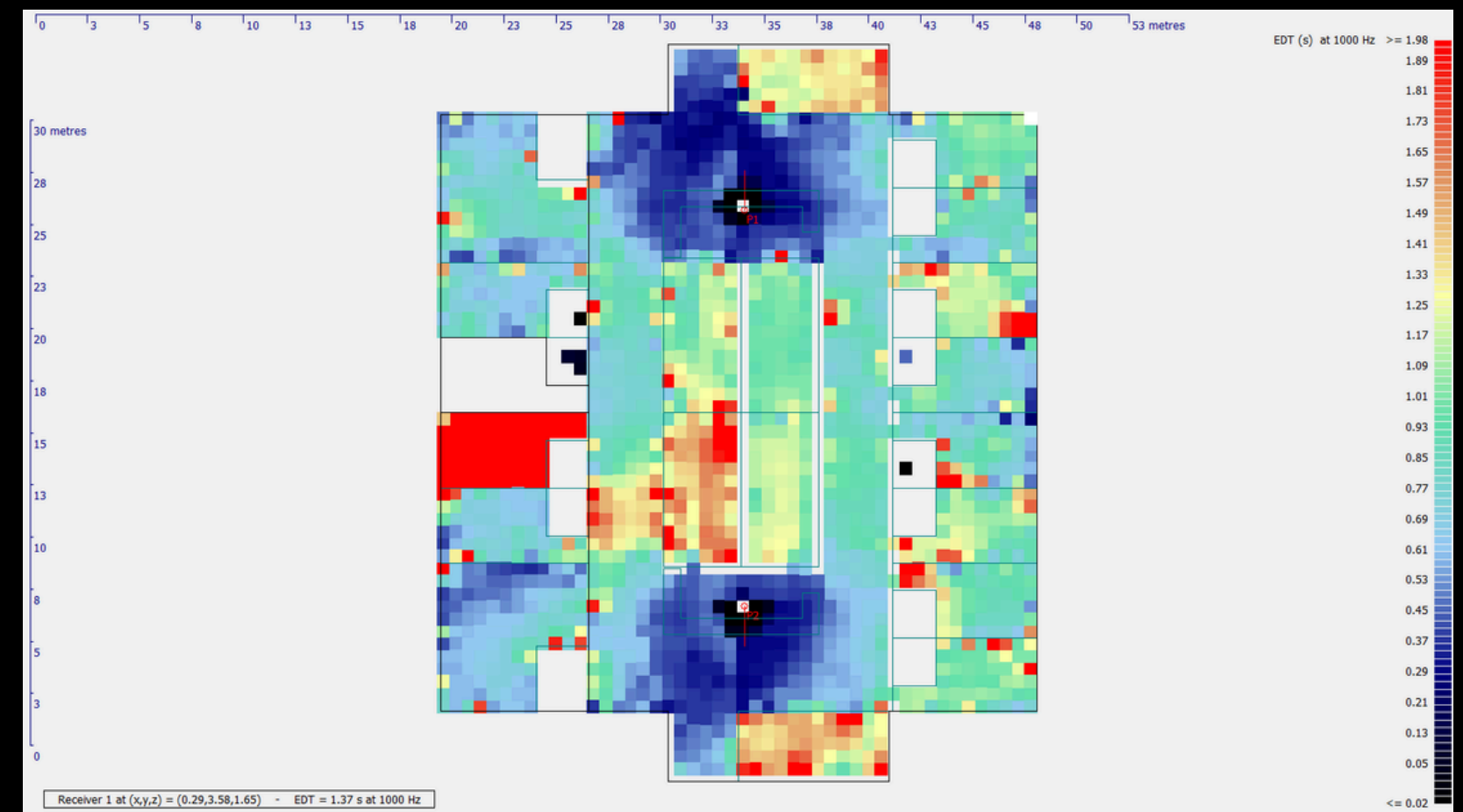
Edt @1000 Hz

The Early Decay Time (EDT) at 1000 Hz remains constant at 1.37 s across both the NC 30 and NC 35 scenarios, confirming that the change in background noise level does not influence the physical decay of sound energy in the room. This specific result at Receiver 1 (simulating a sitting patient at a 1.65 m height) shows a slight increase in reverberance compared to the 500 Hz band, despite the presence of acoustic ceiling tiles. Spatially, the nurse stations continue to show the highest clarity with low EDT values below 0.45 s, while the localized "red zone" in the mid-size ward on the left persists, with EDT values exceeding 1.8 s. This indicates that the ward's architectural configuration and plaster walls are creating consistent acoustic hotspots in the primary speech frequency range, regardless of whether the HVAC system is operating at an NC 30 or NC 35 level.

NC 30



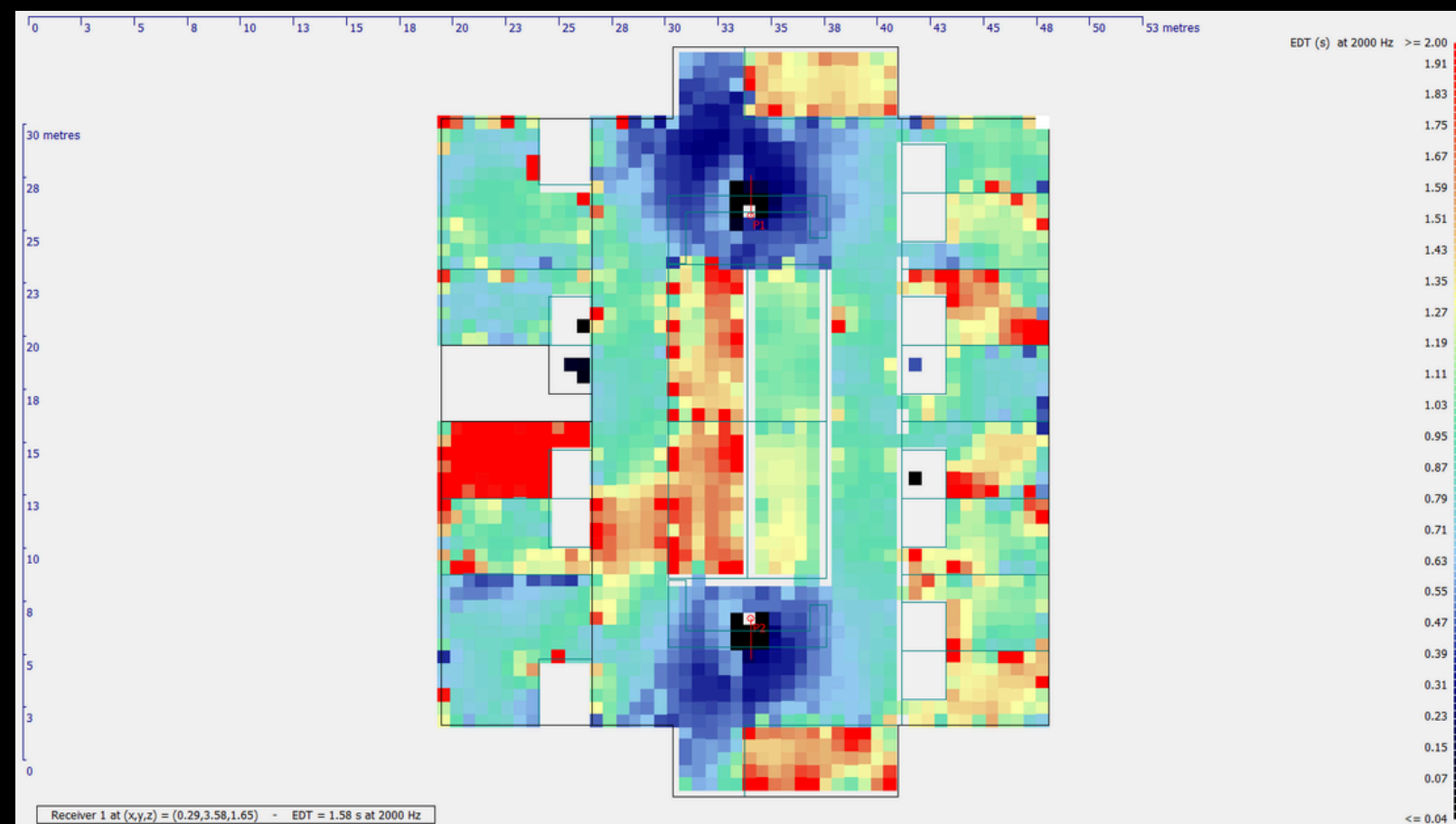
NC 35



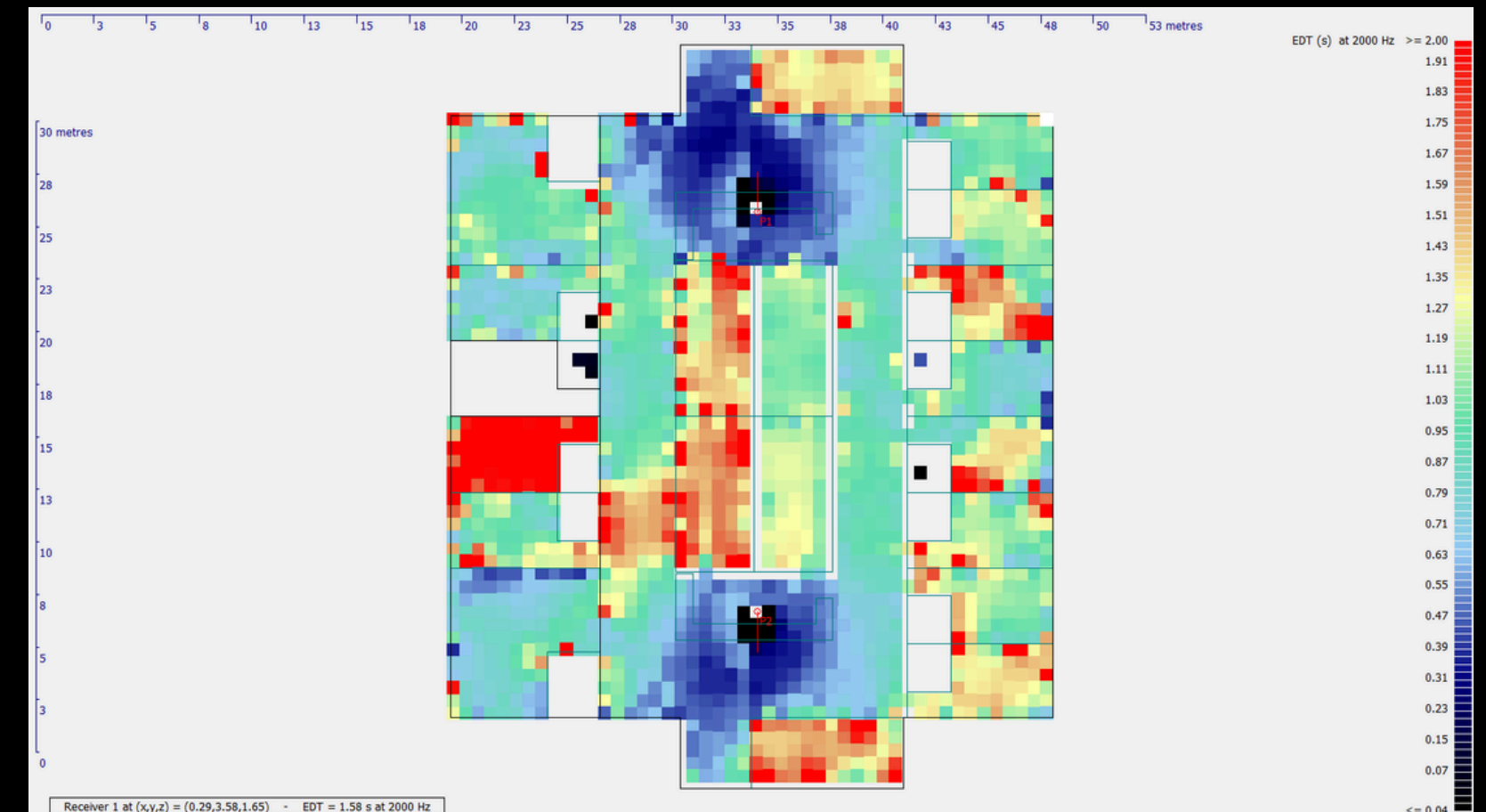
Edt @2000 Hz

At 2000 Hz, the Early Decay Time (EDT) continues to measure identically at 1.58 s for both NC 30 and NC 35 scenarios, as the change in background noise does not alter the physical sound decay within the 2173.00 m³ volume. This result for Receiver 1 (sitting patient at 1.65 m) represents a sharp increase in reverberance compared to the lower bands, aligning with the global reverberation "bump" previously identified at this frequency. Spatially, while the nurse stations remain the most controlled areas with EDT values below 0.47 s, the "red zone" in the left-hand ward expands significantly, with EDT values now consistently exceeding 2.00 s. This indicates that at high speech frequencies—critical for consonant recognition and medical alarm clarity—the combination of plaster walls and the 2.8 m ceiling creates an environment where sound lingers far too long, regardless of the HVAC noise level.

NC 30



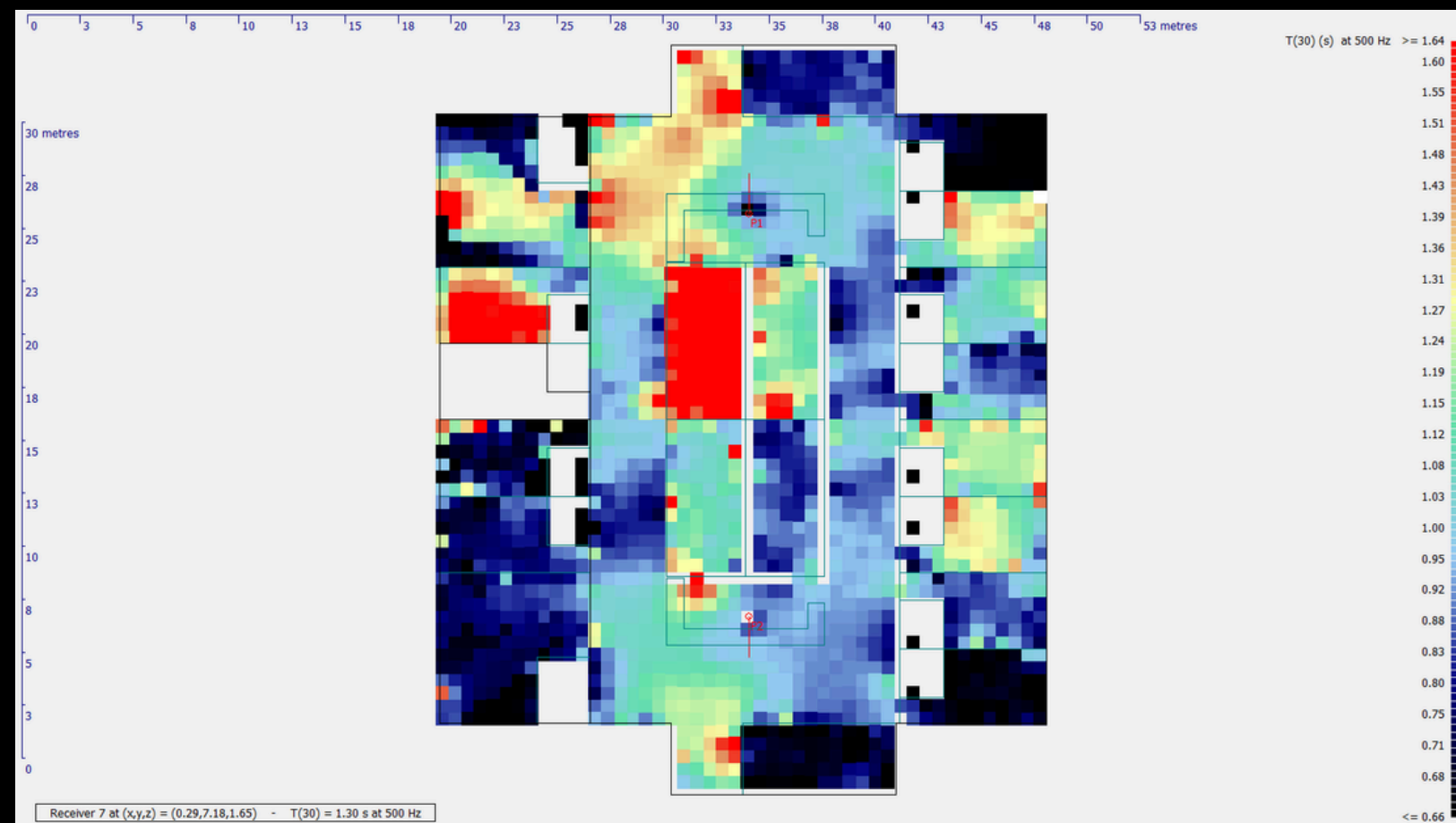
NC 35



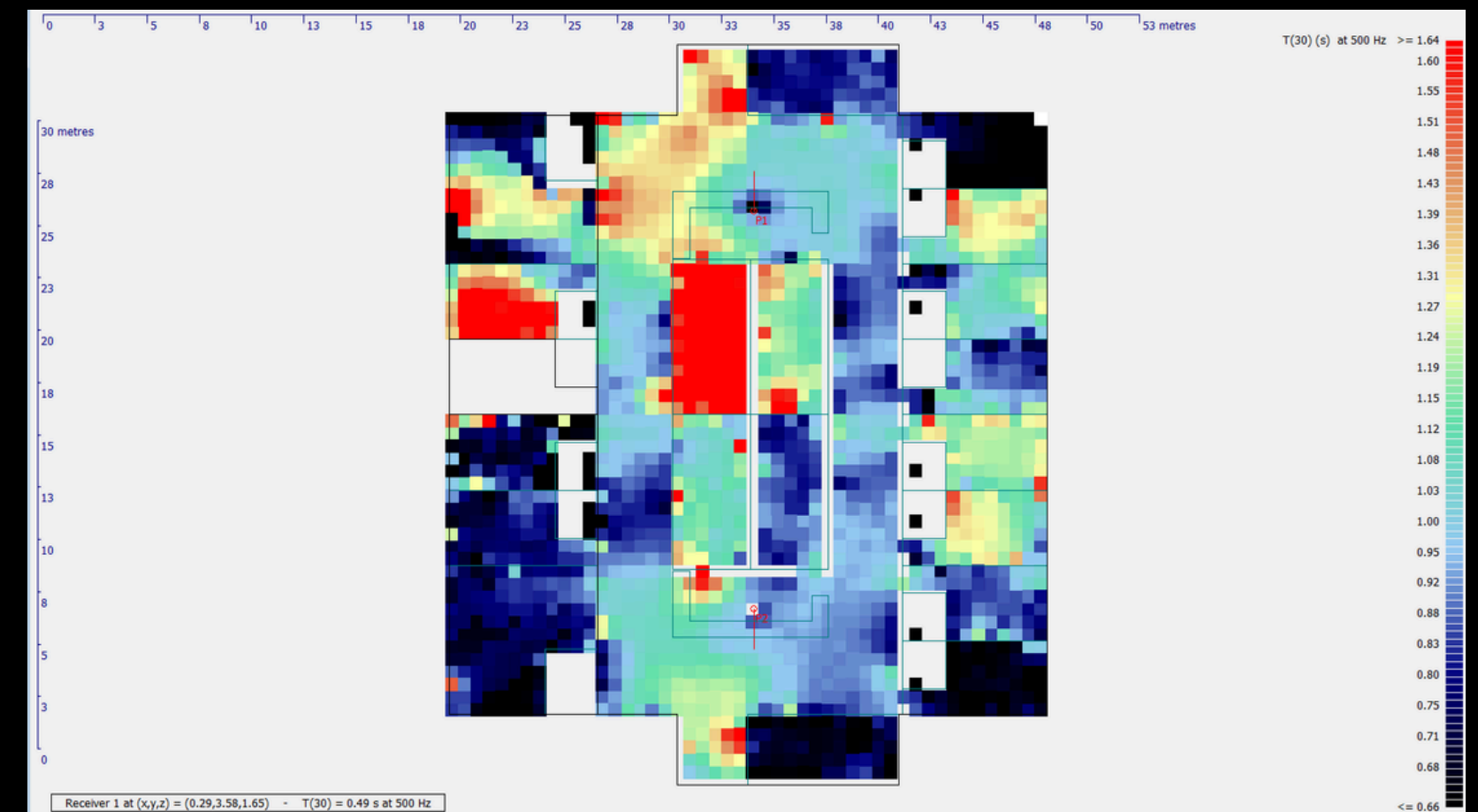
T30 @500 Hz

The Reverberation Time T30 at 500 Hz shows a distinct spatial variance across the ward, though the physical values remain identical between the NC 30 and NC 35 scenarios. At Receiver 1, simulating a patient sitting on a bed at 1.65 m, the measured T30 is 0.49 s, which is significantly lower than the global estimated average of 1.18 s for this frequency band. Spatially, the central service rooms and specific ward areas exhibit a "red zone" of higher reverberation exceeding 1.64 s, likely due to the reflective nature of the plaster walls and vinyl flooring in those localized zones. While the T at the patient's head height is well-controlled at 0.49 s, the high global reverberation suggests that background noise will still accumulate and persist within the 2173.00 m³ volume, potentially increasing the cognitive load on medical staff regardless of the HVAC noise criteria.

NC 30



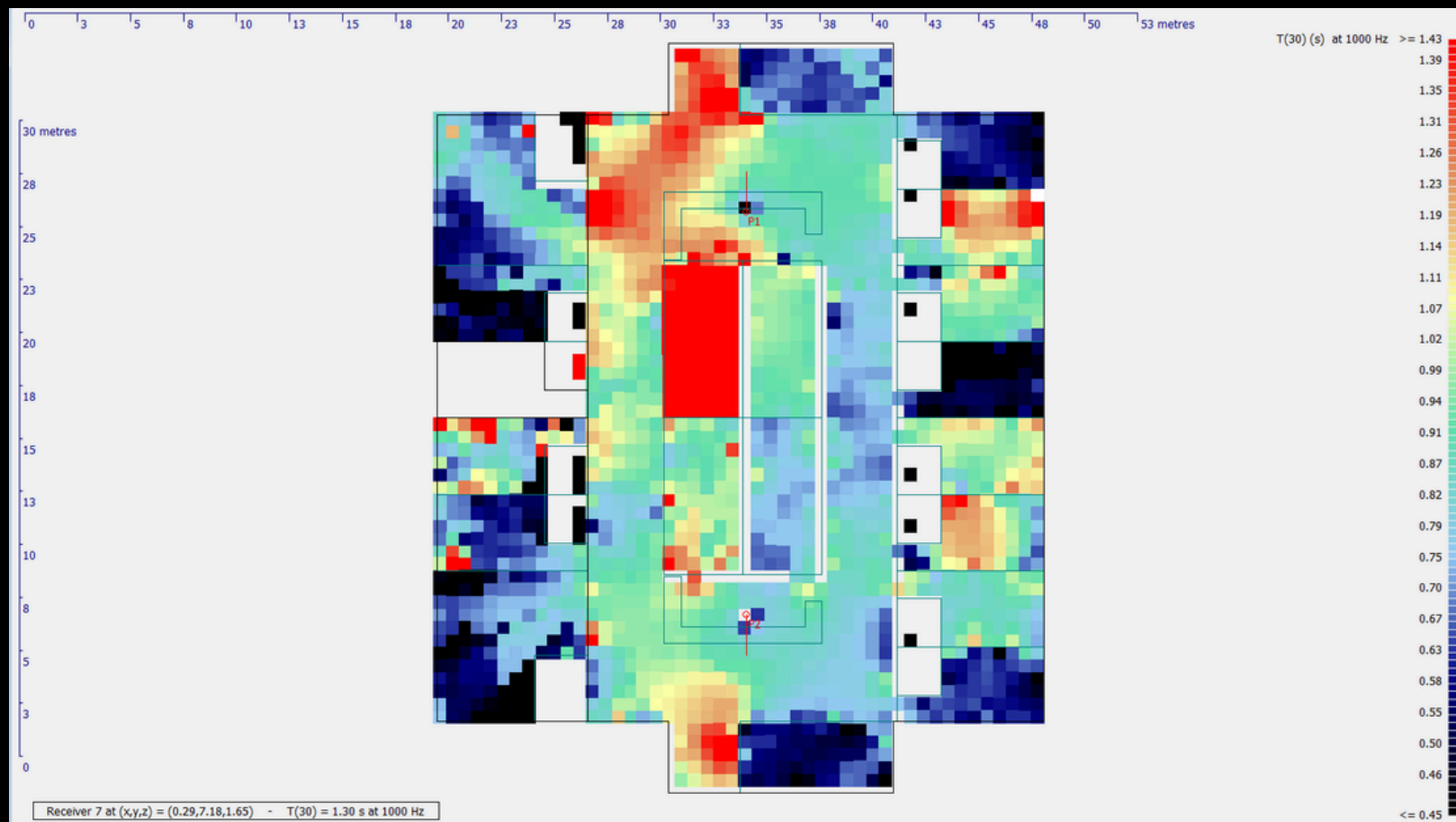
NC 35



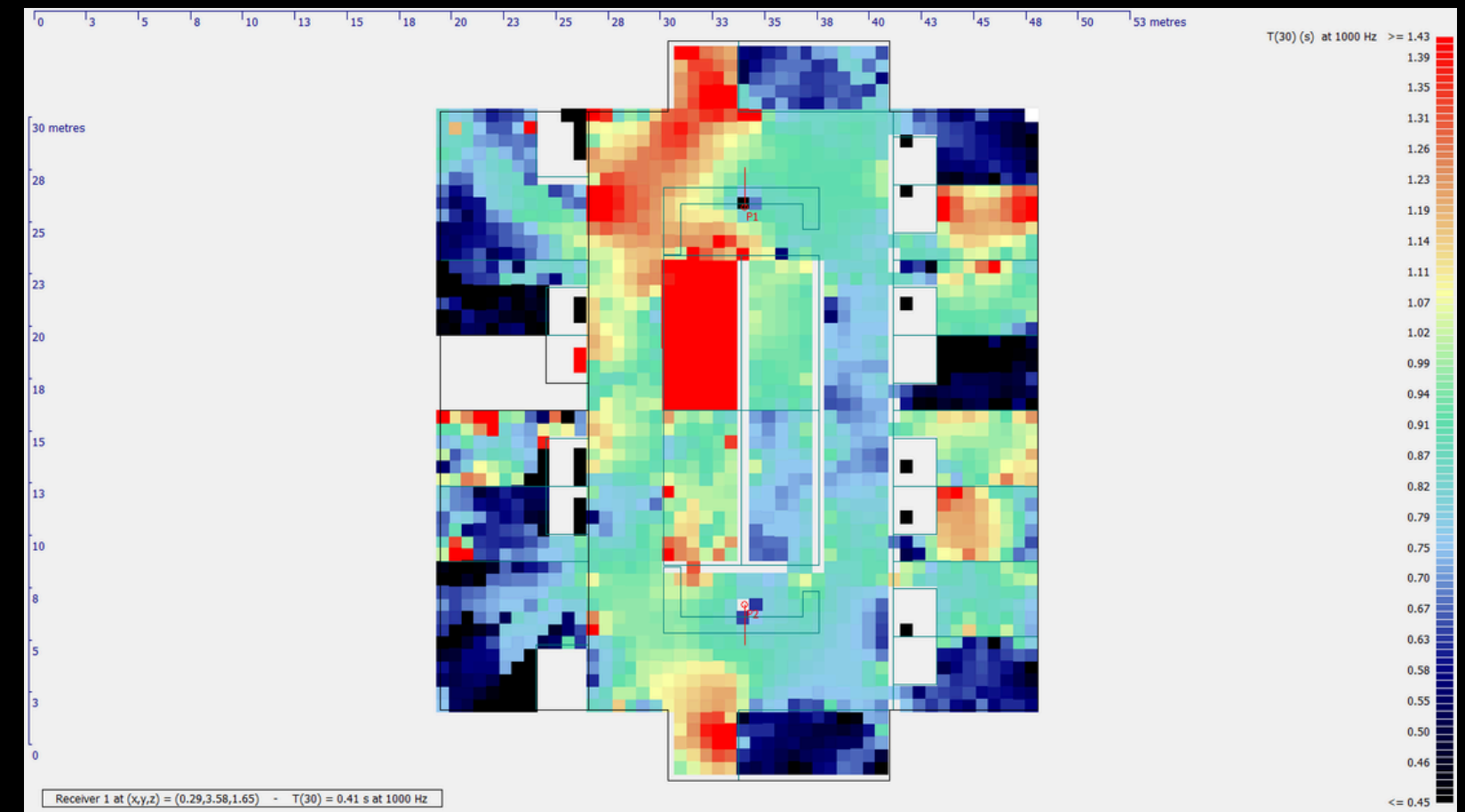
T30 @1000 Hz

The Reverberation Time T30 at 1000 Hz remains physically identical across both NC 30 and NC 35 scenarios, as the background noise floor does not alter the energy decay rates of the ward. At Receiver 1, simulating a patient sitting at a 1.65 m height, the measured T30 is 0.41 s, representing a slight improvement in localized control compared to the 500 Hz band. This localized result is significantly better than the global estimated average of 1.09 s at this frequency, which is where the acoustic ceiling tiles are expected to be most efficient. Spatially, the central service rooms continue to exhibit a high-reverberation "red zone" with values exceeding 1.43 s, likely due to the lack of absorptive materials in these high-traffic areas. While the patient's immediate environment is acoustically "dry," the high global reverberation across the 2173.00 m³ volume indicates that noise from these service zones will likely persist and propagate throughout the corridors.

NC 30



NC 35

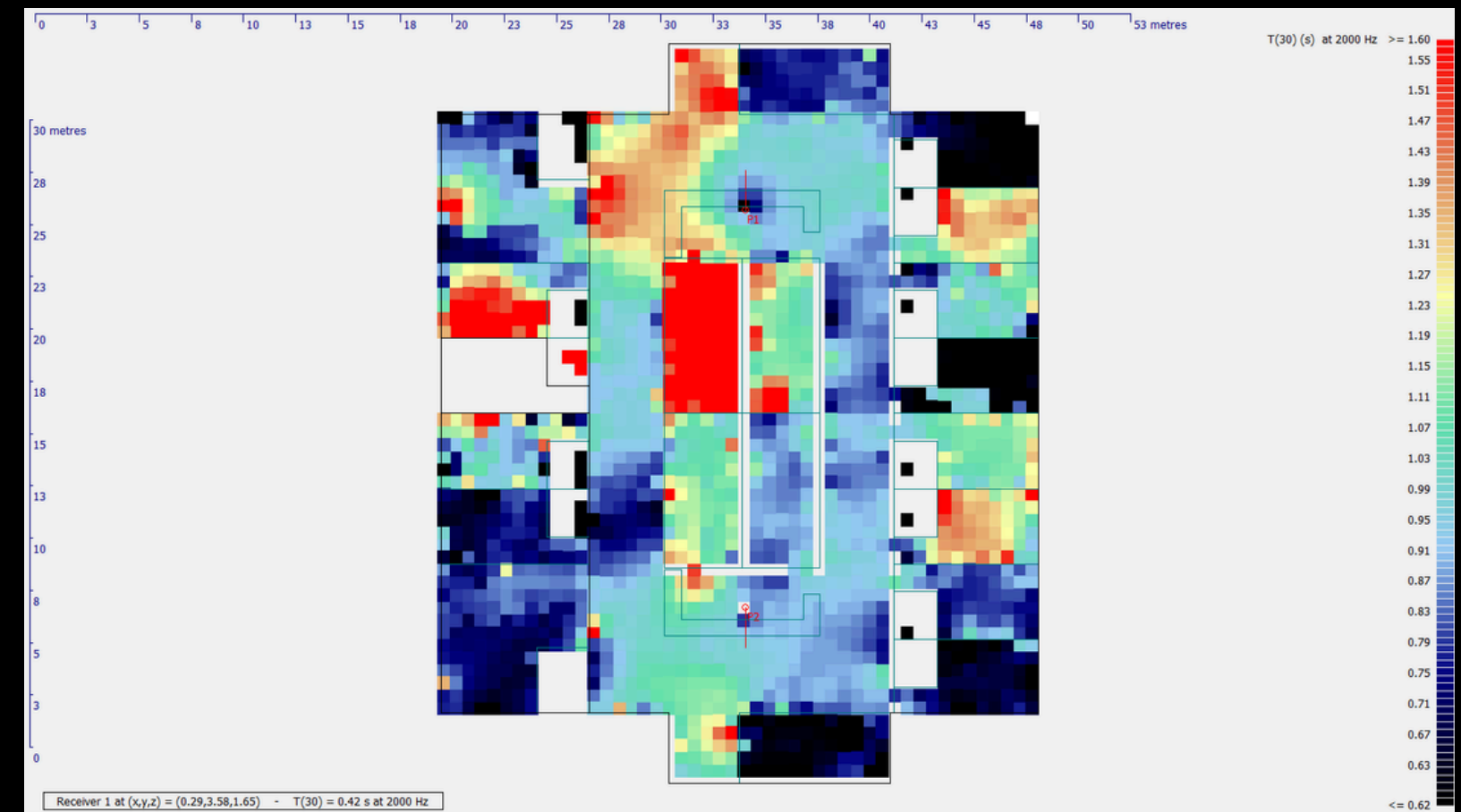
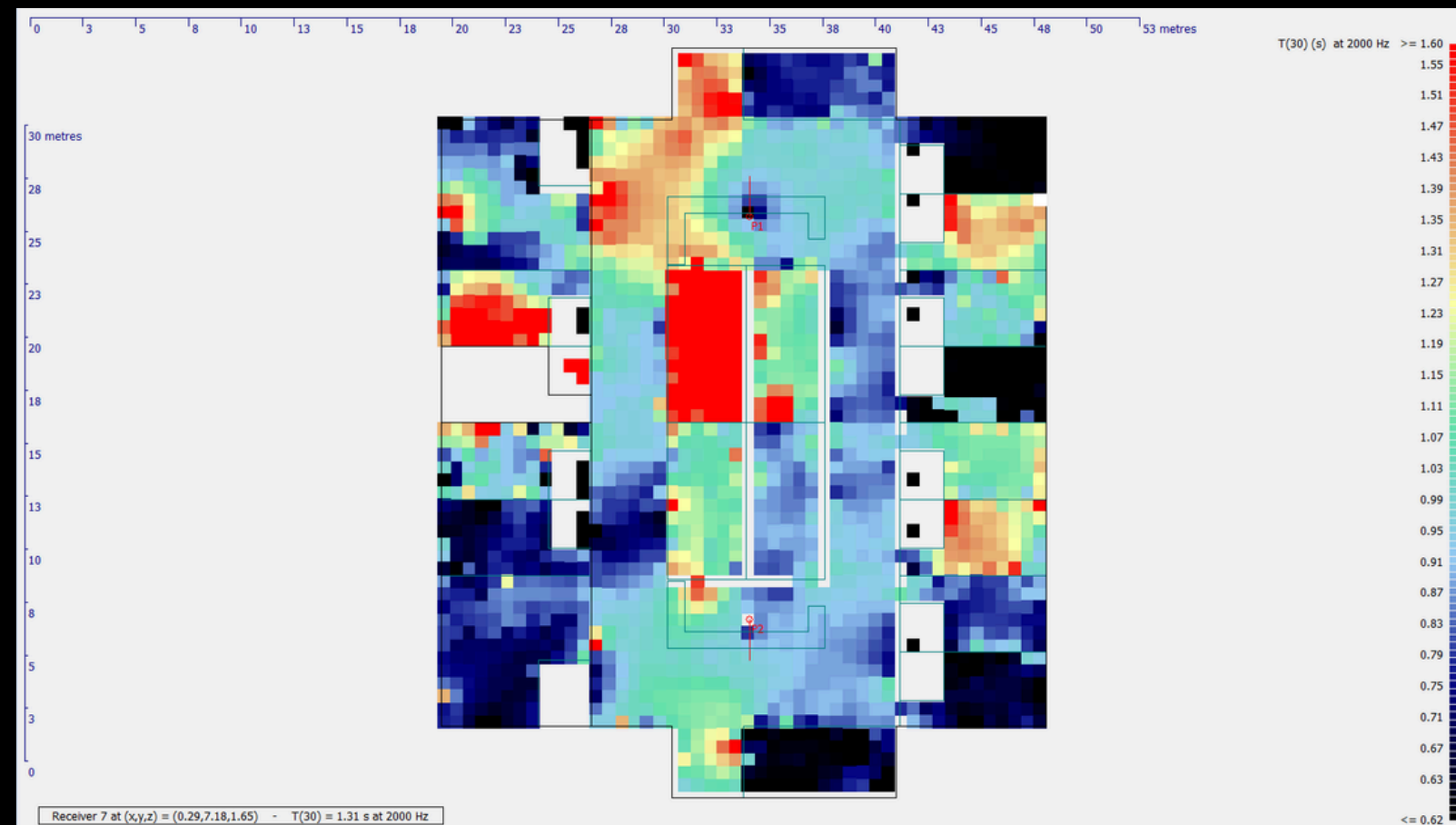


T30 @2000 Hz

The Reverberation Time T30 at 2000 Hz remains physically identical across both the NC 30 and NC 35 scenarios, as the change in the HVAC noise floor does not affect the energy decay rates of the ward. At Receiver 1 (sitting patient at 1.65 m), the measured T30 is 0.42 s, indicating that the localized acoustic environment for the patient remains well-controlled even during the high-frequency reverberation "bump". This local result is significantly lower than the global estimated average of 1.18 s for this frequency band. Spatially, the central service rooms continue to show a "red zone" of higher reverberation exceeding 1.60 s, suggesting that the plaster walls and lack of high-frequency absorption in these zones create hotspots for sound persistence. While the patient's immediate area is acoustically "dry," the elevated global reverberation at 2000 Hz means that high-pitched noises, such as medical alarms, will likely linger and propagate through the corridors.

NC 30

NC 35

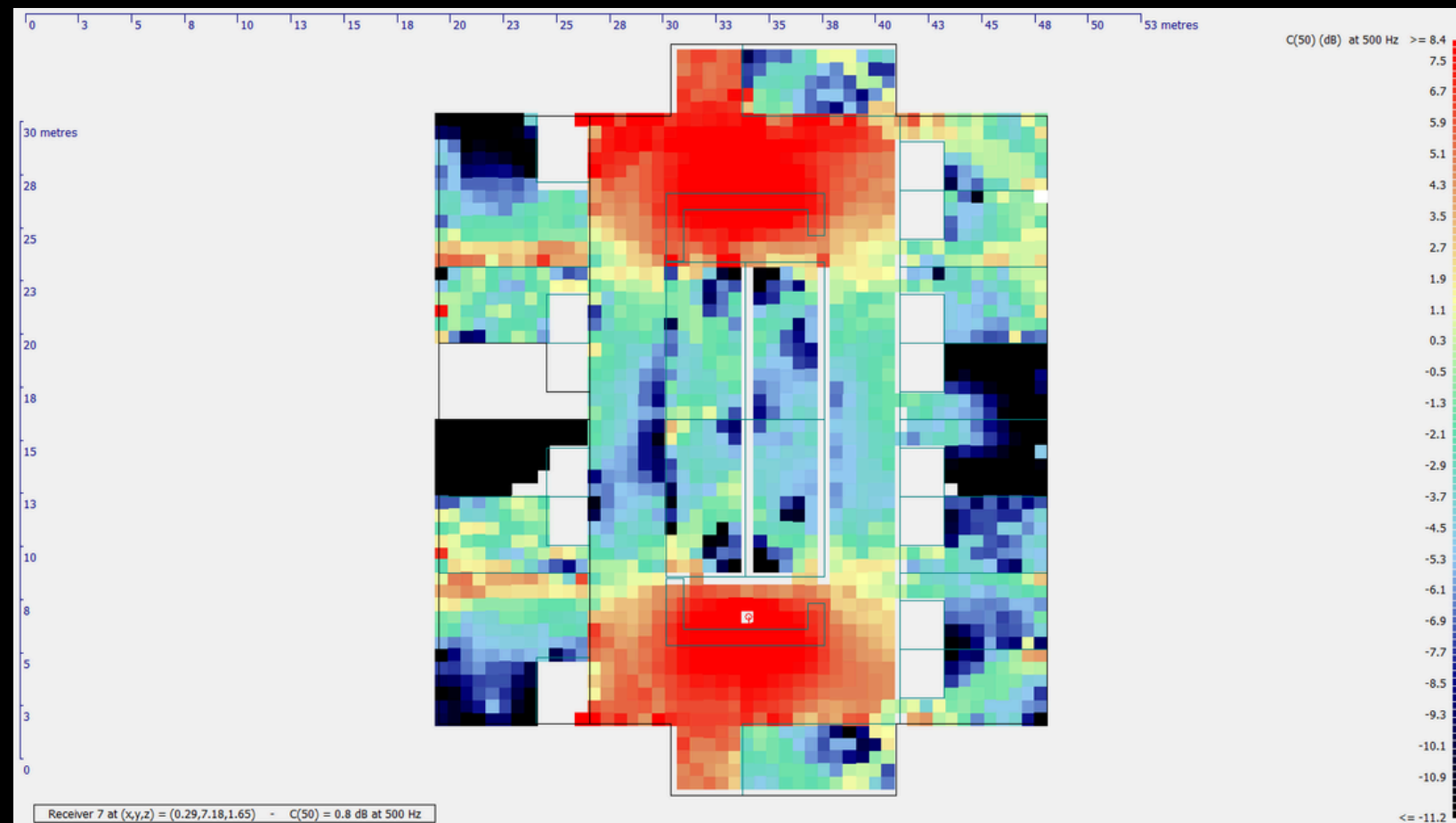


C50 @500 Hz

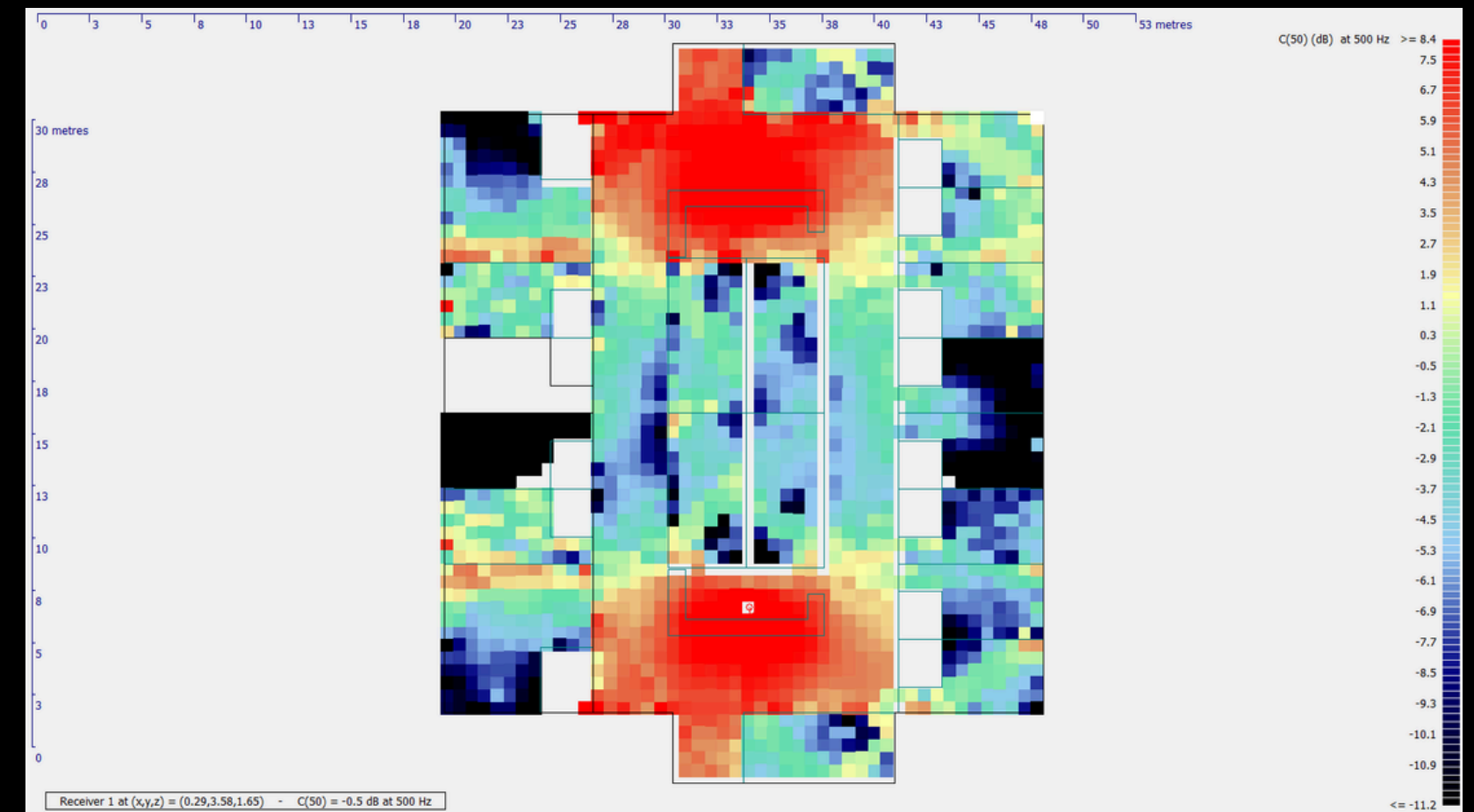
The Clarity C50 at 500 Hz is the first parameter in this analysis that demonstrates a direct sensitivity to the change in background noise criteria. While the physical reverberation times remained constant, the shift in the noise floor causes a measurable degradation in speech clarity due to a reduced Signal-to-Noise Ratio (SNR).

- Receiver 1 Performance: For a patient sitting at 1.65 m, the C50 drops from 0.8 dB at NC 30 to -0.5 dB at NC 35. This transition into a negative decibel value indicates that the late (reflected) energy and background noise are now more dominant than the direct speech signal, which is a significant clinical concern for verbal communication at the bedside.
- Spatial "Clear" Zones: The nurse stations (central red zones) maintain high clarity above 8.4 dB in both scenarios, indicating that the immediate architectural layout in these hubs effectively prioritizes direct sound.
- Spatial "Dead" Zones: Large portions of the peripheral ward rooms and the central patient services corridor show negative C50 values (blue/black regions), with some areas dropping below -11.2 dB. The expansion of these low-clarity zones in the NC 35 map confirms that even a 5 dB increase in HVAC noise significantly broadens the areas where speech is likely to be perceived as "muddled" or unintelligible by both patients and staff.

NC 30



NC 35

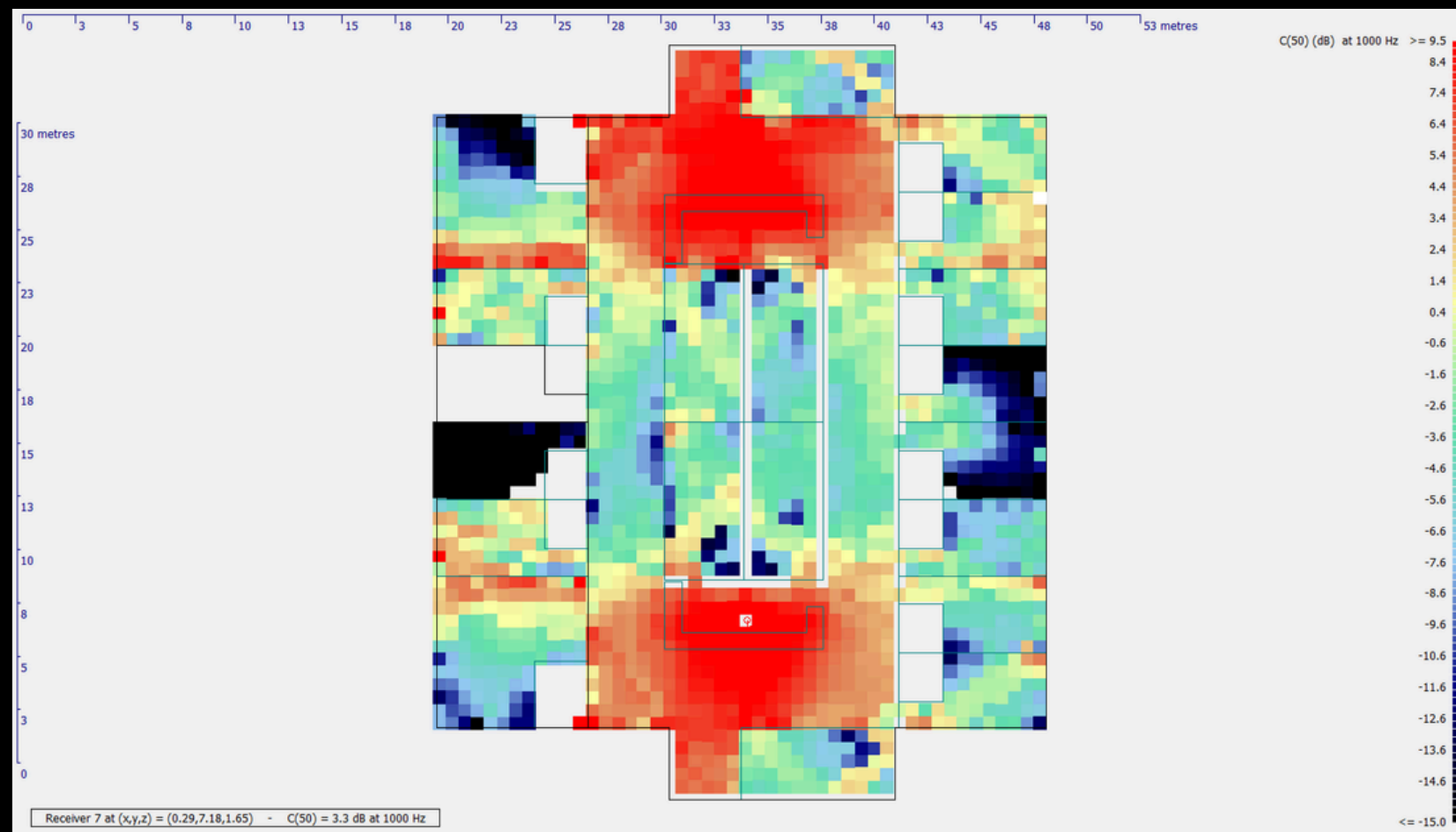


C50 @1000 Hz

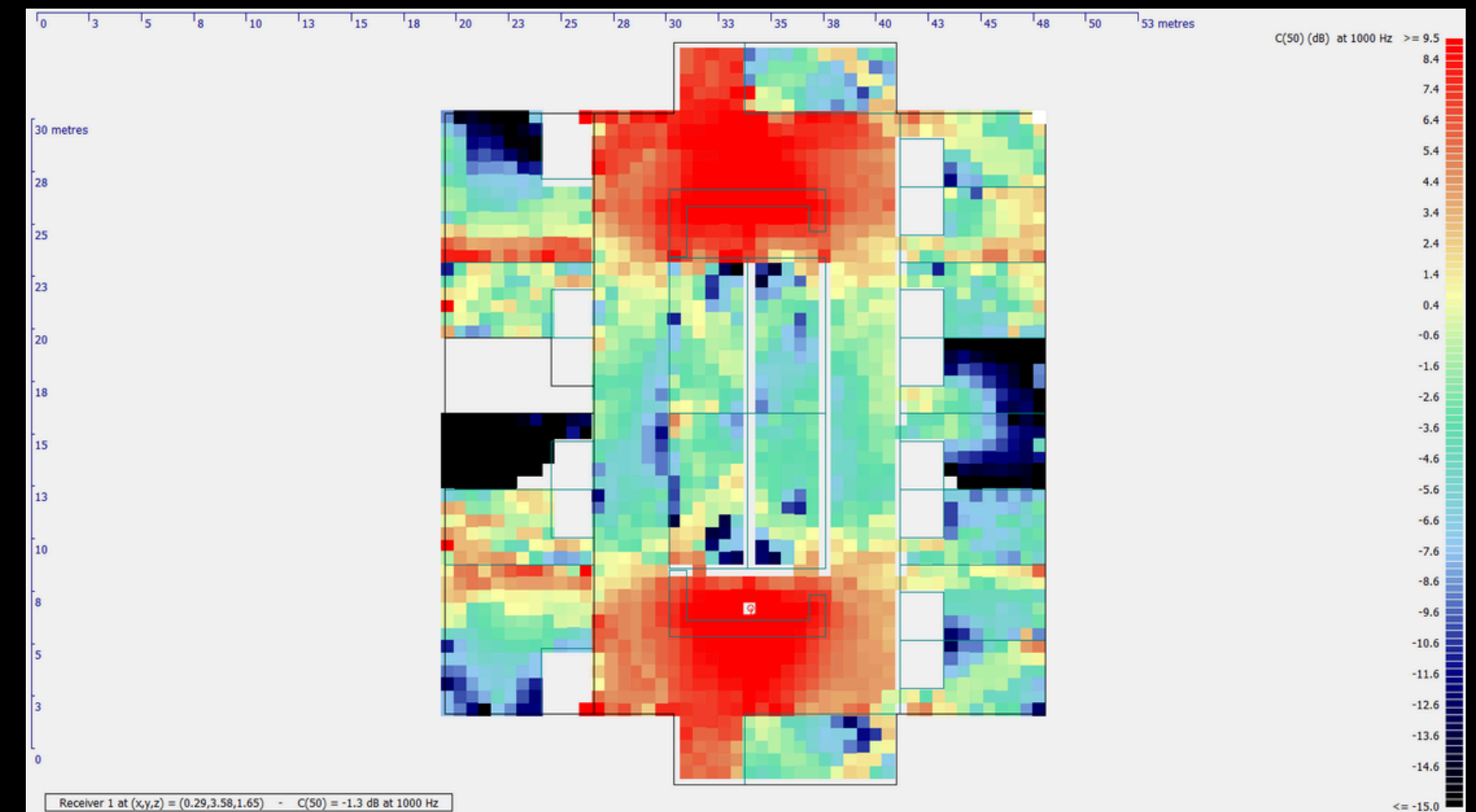
The analysis of Clarity C50 at 1000 Hz reveals a critical decline in acoustic performance when transitioning from a high-performance noise environment to a standard one.

- Receiver 1 Performance: At the patient's position (1.65 m height), the C50 drops significantly from 3.3 dB at NC 30 to -1.3 dB at NC 35. This shift from a positive to a negative decibel value at this frequency—the peak of the human speech spectrum—indicates that background HVAC noise and late reflections now overpower the direct sound of speech.
- Spatial Distribution: The nurse stations (central red zones) maintain high clarity levels above 9.5 dB, ensuring that critical communication between staff remains reliable in both scenarios.
- Clinical Risk Zones: The peripheral ward rooms and the central service corridor exhibit an expansion of "dead zones" (dark blue/black), with clarity values plummeting to -15.0 dB. The prevalence of these negative values at NC 35 suggests that even if the patient's local area is acoustically "dry," the background noise level is high enough to significantly blur clinical instructions and alarms in this primary speech band.

NC 30



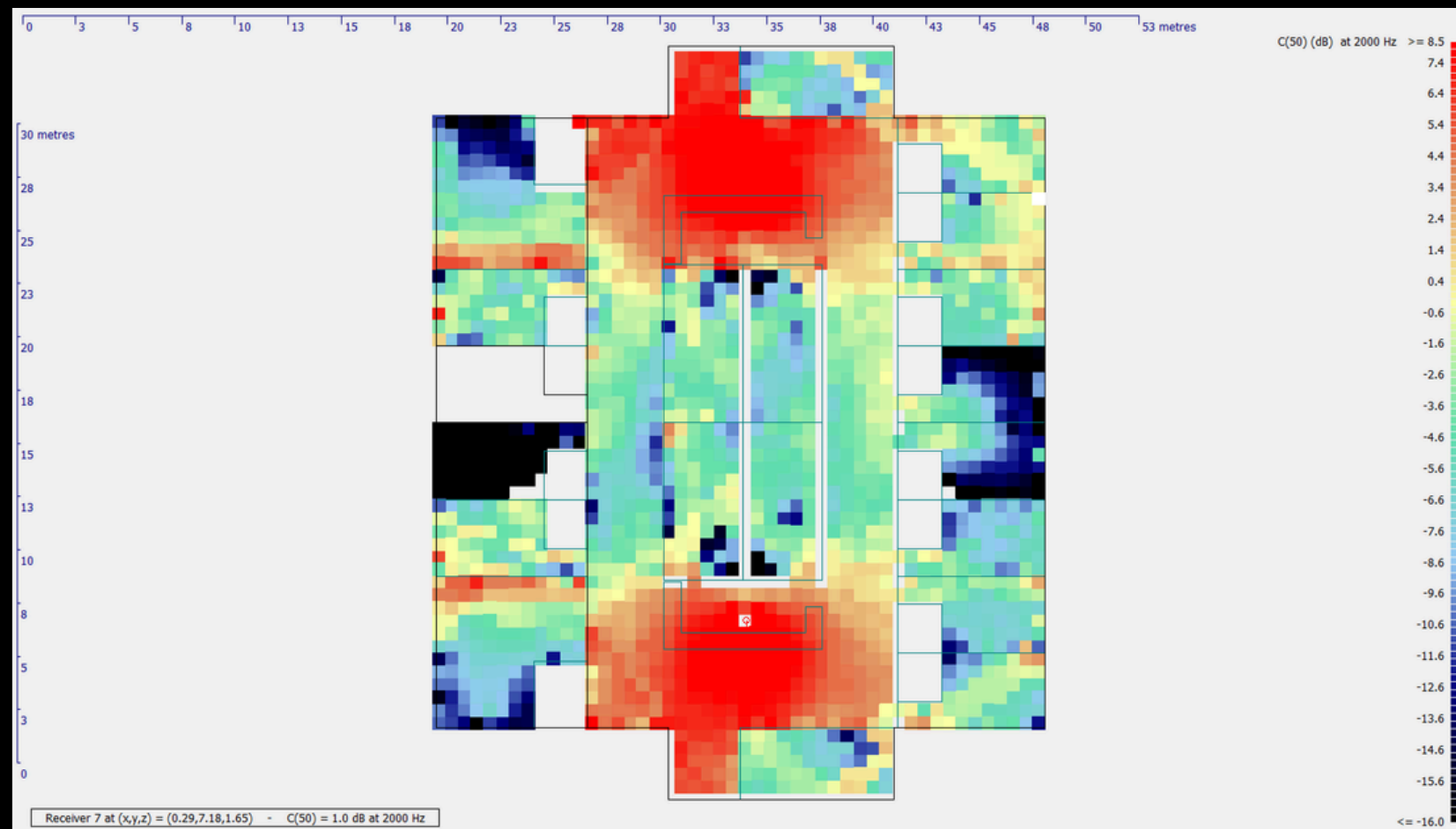
NC 35



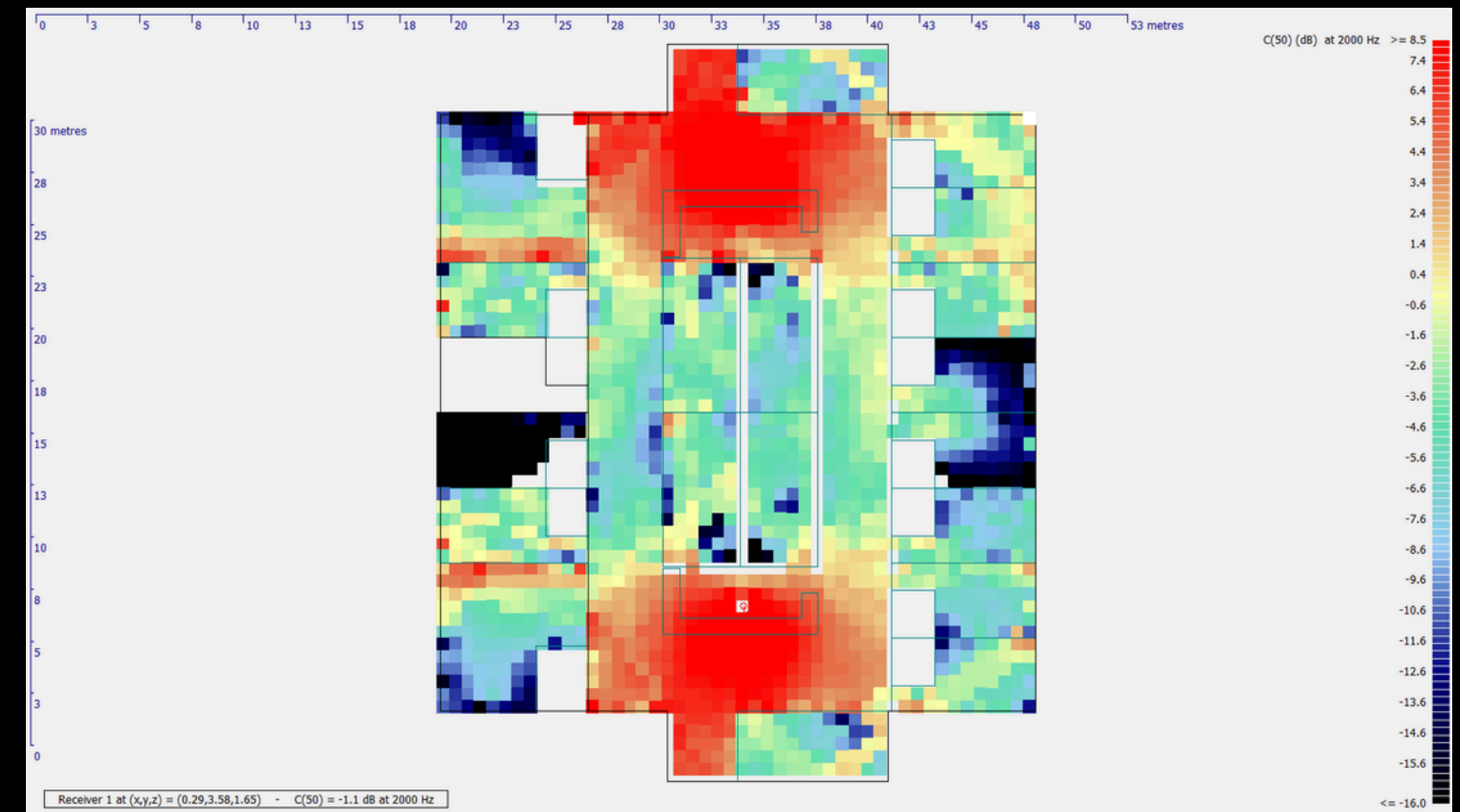
C50 @2000 Hz

The Clarity C50 at 2000 Hz shows a critical performance drop at Receiver 1 (simulating a sitting patient at 1.65 m), falling from 1.0 dB at NC 30 to -1.1 dB at NC 35. This transition into negative decibel values at 2000 Hz is particularly concerning for clinical safety, as this high-frequency band is vital for consonant recognition and the distinctness of medical alerts. Spatially, the nurse stations remain the most intelligible areas with clarity values maintained above 8.5 dB, while the "dead zones" in the peripheral ward rooms and central service corridors expand significantly, with C50 values plummeting as low as -16.0 dB. This confirms that the high-frequency reverberation "bump" previously identified, when combined with a standard NC 35 HVAC noise floor, effectively masks critical sound information, making the ward environment significantly less intelligible for both patients and staff.

NC 30



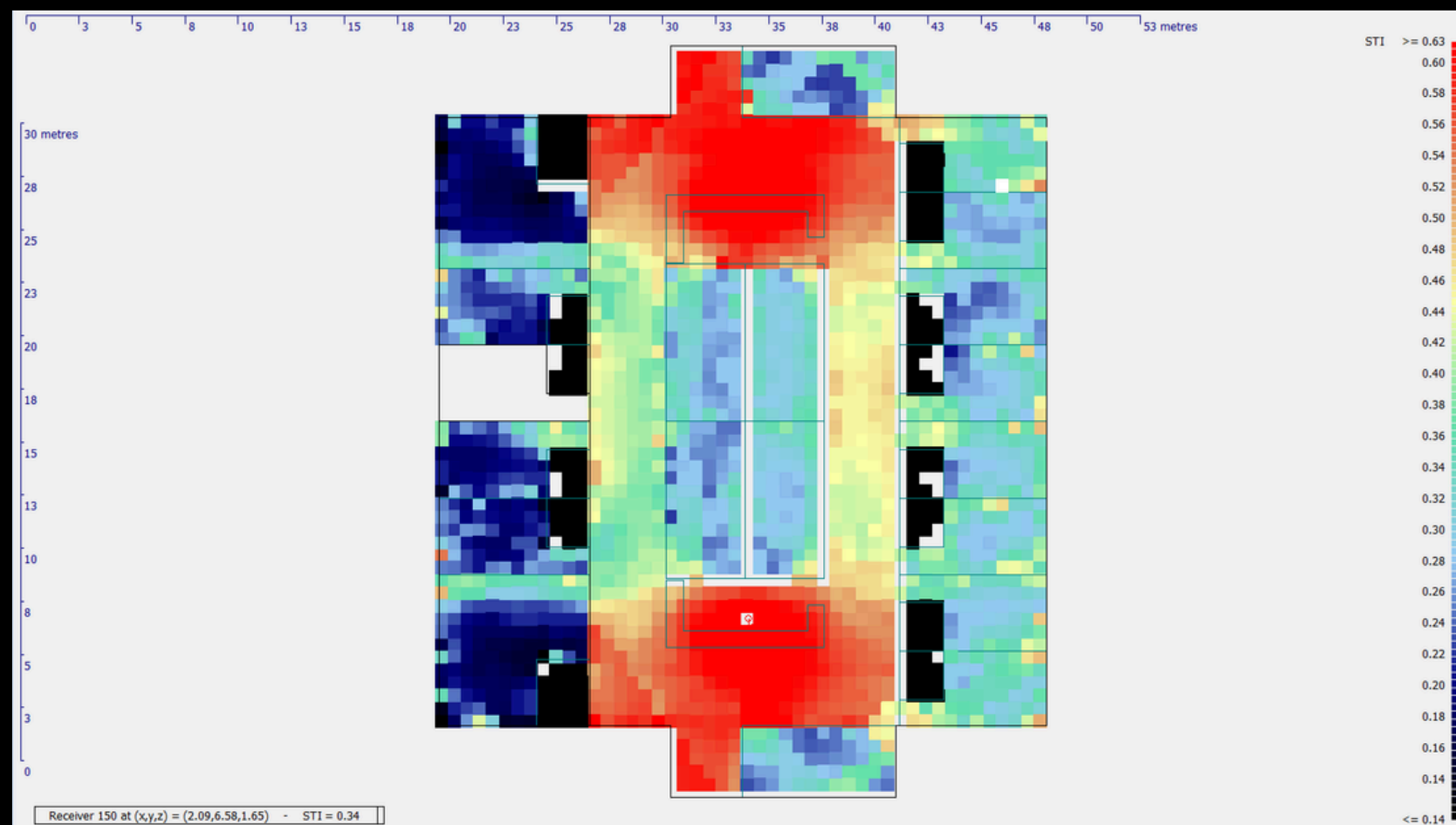
NC 35



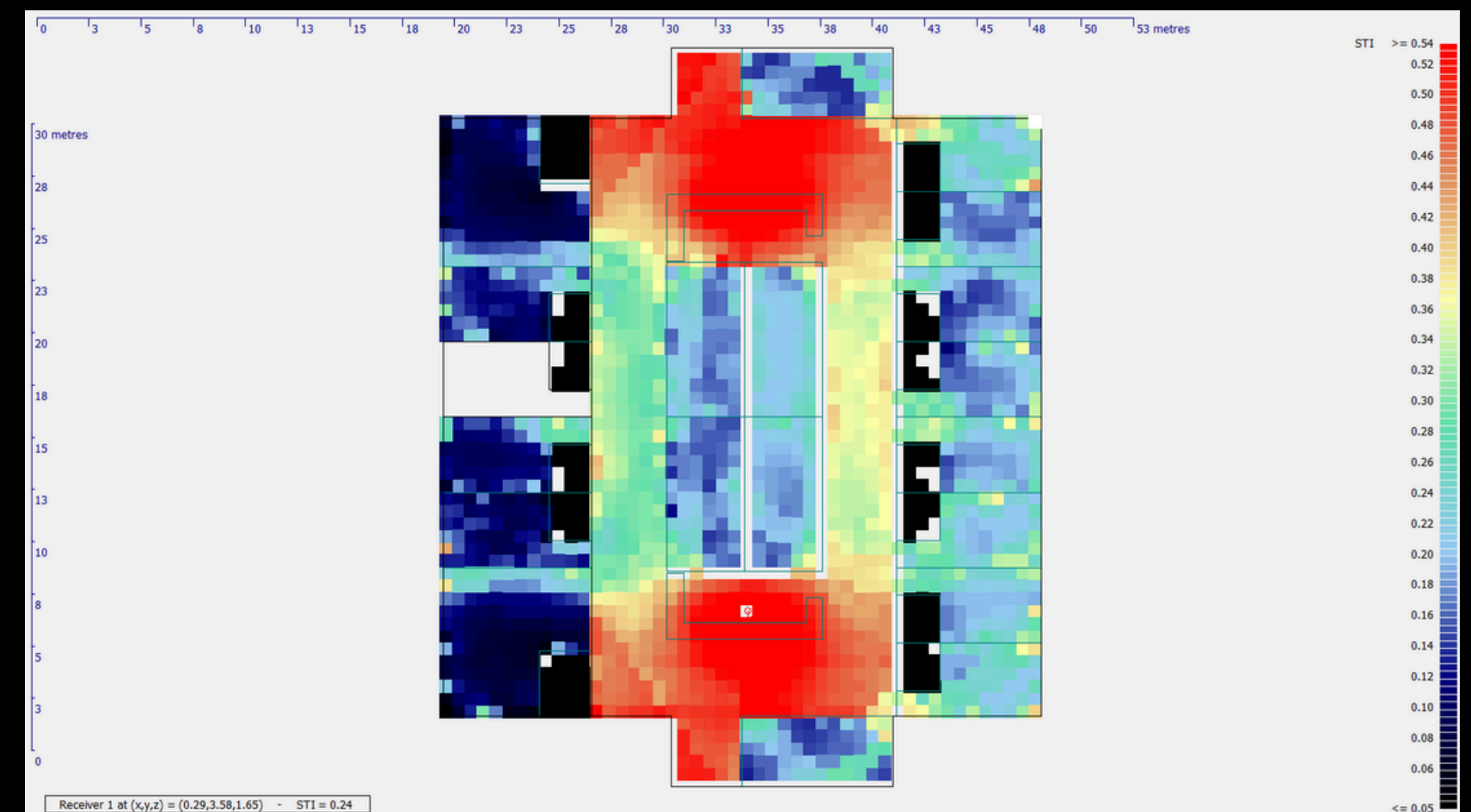
STI

The Speech Transmission Index (STI) results demonstrate a severe degradation in overall intelligibility as the background noise floor shifts from NC 30 to NC 35. In the NC 30 scenario, the highest STI values reach ≥ 0.63 at the central nurse stations, indicating "Good" to "Excellent" intelligibility in those hubs, while a specific receiver point shows an STI of 0.34, which is categorized as "Poor". Upon increasing the noise level to NC 35, the maximum STI drops significantly to ≥ 0.54 , and the patient position at Receiver 1 (1.65 m height) plummets to a nearly unintelligible value of 0.24. Spatially, the red zones representing high intelligibility around the nurse stations shrink considerably, while the "dead zones" expand throughout the peripheral ward rooms, with STI values falling as low as 0.05. This confirms that while the ward's architectural layout—characterized by its 2.8 m clear height and plaster walls—is already acoustically challenging, the introduction of NC 35 HVAC noise reduces speech transmission to critical levels across the majority of the 2173.00 m³ volume.

NC 30



NC 35



Conclusion and Clinical Recommendations

The comparative analysis of the 2173.00 m³ patient ward demonstrates that background noise levels are the primary driver of speech intelligibility, as the room's physical decay remains constant.

1. Summary of Findings

- **Intelligibility Collapse:** Transitioning to NC 35 reduces the STI at the patient's bedside (Receiver 1) to 0.24, a level considered "Unintelligible" for critical clinical communication.
- **Negative Clarity (C50):** At NC 35, clarity values at the patient's head height drop below 0 dB across all core frequencies (500Hz, 1000Hz, and 2000Hz), meaning reflected energy and background noise successfully mask direct speech.
- **Frequency Vulnerability:** The global reverberation "bump" at 2000 Hz (1.18 s) coincides with the frequency of most medical alarms, creating a high-risk environment for alarm fatigue and missed signals when combined with an NC 35 floor.
- **Spatial Variance:** While the nurse stations maintain localized control, the "dead zones" in peripheral wards expand significantly at NC 35, compromising staff-to-patient communication safety.

2. Clinical Recommendations

- **HVAC Calibration:** The mechanical system must be restricted to NC 30 or lower to maintain a baseline of "Fair" intelligibility for recovering patients.
- **Targeted Absorption:** Additional acoustic treatment is required in the left-side ward "red zones" where EDT exceeds 1.8 s, likely due to parallel reflective plaster walls.
- **Ceiling Optimization:** Given the 2.8 m clear height, high-performance acoustic tiles with a high Noise Reduction Coefficient (NRC) at 2000 Hz should be prioritized to flatten the high-frequency reverberation bump.
- **Operational Strategy:** Since doors were modeled as transparent, actual operational safety will rely on keeping ward doors closed to mitigate "Acoustic Coupling" and noise spill from service corridors.

References

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